

Robust Aggregation of Expert Probability Forecasts via Wasserstein Barycenters

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Abstract

We study the aggregation of expert probability forecasts over a finite outcome space. The standard approach, the arithmetic mean (linear opinion pool), is sensitive to extreme opinions: a minority assigning substantial mass to tail outcomes can distort the aggregate away from the majority view. We propose the Wasserstein barycenter as a robust alternative. Under the indicator ground cost the barycenter reduces to an ℓ^1 aggregation problem; for ordered outcome spaces it admits a closed form: the distribution whose CDF is the component-wise median of the individual CDFs, computable without a solver. We validate the framework across three datasets. On 94 binary and 113 multi-category questions from the Good Judgment Project (GJP, Years 1–4), the ℓ^1 barycenter reduces mean Brier score by 12.5% on binary questions ($p < 0.001$) and 8.7% on multi-category questions with $K \in \{3, 4, 5\}$ outcomes ($p < 0.001$); the multi-category reduction grows with K , reaching 17.2% for $K = 5$. On 76 quarters of U.S. Survey of Professional Forecasters core CPI histograms (2007–2025), benchmarked against the arithmetic mean, logarithmic pool, and trimmed mean, the barycenter and trimmed mean both achieve lower mean RPS than the arithmetic mean (Wilcoxon $p = 0.005$;

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DM $p = 0.23$, the latter reflecting variance inflation from the 2021–2022 inflation surge), with the advantage concentrated in low-dispersion quarters (DM $p < 0.001$); the logarithmic pool has the highest P(correct bin) but worst RPS, a divergence explained by the ℓ^1 – ℓ^2 mismatch between the pooling objective and the scoring rule. A replication on the ECB Survey of Professional Forecasters confirms the dispersion-conditional pattern on an independent Euro-area panel. Simulation evidence characterizes the robustness–efficiency tradeoff and identifies conditions under which each method is preferred. For applied use, we recommend a dispersion-adaptive convex blend of the barycenter and the arithmetic mean: estimated on an expanding historical window, it achieves the lowest out-of-sample RPS of all methods evaluated, improving over the arithmetic mean by 4.5%, and automatically adapts to the panel’s cross-sectional agreement without manual tuning.

Keywords: forecast combination; opinion pooling; Wasserstein distance; optimal transport; Survey of Professional Forecasters; robust aggregation

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1 Introduction

Combining probabilistic forecasts from multiple experts is a central problem in macroeconomics, finance, and public policy. The Survey of Professional Forecasters (SPF) elicits probability distributions over future GDP growth, inflation, and unemployment from panels of professional economists every quarter; these individual histogram forecasts must be summarized into a single aggregate distribution before informing decisions. Expert elicitation in public health, climate science, and risk analysis faces the same aggregation problem. In all of these settings the data consist of probability vectors over a finite outcome space (histogram bins, policy categories, or categorical scenarios), and the goal is to produce a single representative probability distribution.

The dominant method is the *linear opinion pool*, the arithmetic mean of the individual probability vectors [15, 17, 21]. Its appeal is axiomatic: it is the only combination rule consistent with marginalisation across subsets of outcomes [15]. In well-behaved panels with exchangeable, well-calibrated forecasters it is also asymptotically efficient. Its practical weakness is sensitivity to heterogeneous or extreme opinions. A single forecaster who places substantial mass on an extreme outcome shifts the aggregate by $1/N$ of that distance, regardless of how far outside the consensus view that forecaster sits. In the SPF, some forecasters routinely assign 30–50% probability to tail scenarios that the median forecaster assigns near zero; the arithmetic mean faithfully imports these extreme probabilities into the aggregate.

The forecast combination literature has proposed a range of alternatives to the arithmetic mean [4, 21]. The *trimmed mean* removes the most extreme forecasters before averaging; it requires choosing a trim fraction and its extension to K -dimensional histogram forecasts is not geometrically principled. The *logarithmic opinion pool* shrinks extreme probabilities by taking the geometric mean, renormalized [15]; in our evaluation it achieves the highest probability mass on the realized bin but the worst Ranked Probability Score (RPS), a divergence we explain analytically via a geometric mismatch between the log pool’s mode-seeking objective and the squared-CDF structure of RPS. Recalibration methods [5, 6] address a distinct

objective: restoring calibration that the linear pool destroys even when individual forecasters are calibrated. Our objective is forecast accuracy, and we neither claim calibration for the barycenter nor optimize for it. Combination methods that minimize expected proper scoring-rule loss [16] require historical accuracy distributions for individual forecasters; our approach requires no track record. Our contribution is a parameter-free, geometry-based aggregation rule whose properties are analytically characterizable for finite categorical outcome spaces (histogram forecasts, policy surveys, and categorical expert elicitations) and whose regime-conditional advantages are theoretically predicted rather than observed post hoc.

Relation to existing work

Trimmed mean. The trimmed mean removes the $\lfloor \alpha N \rfloor$ most extreme forecasters at each tail and averages the remainder [7]. In our US SPF evaluation it achieves similar RPS to the barycenter (the gap is statistically insignificant by both Wilcoxon $p = 0.82$ and DM $p = 0.63$ tests), but it has two limitations our approach avoids. First, α must be chosen; the choice is consequential and there is no principled default for K -dimensional histogram forecasts, where “extreme” is not well-defined without a scalar ordering. Second, trimming provides no geometric interpretation of why certain opinions are downweighted or when the downweighting helps. The barycenter is parameter-free and its regime-conditional advantage is predictable from the closed-form theory (Corollary 2.8).

Logarithmic opinion pool. The log pool concentrates mass near the geometric mean of the individual densities, which is mode-seeking and aggressive: near-zero probabilities collapse toward zero [15], and extremizing [20] amplifies this further. In our evaluation the LP achieves the highest probability on the realized bin but the worst RPS, a divergence we explain analytically via the ℓ^1 – ℓ^2 mismatch of Remark 2.9: mode-seeking reduces ℓ^1 error at the modal bin but increases squared CDF deviations in quarters where a non-modal bin is realized.

Recalibration. [5] show the linear pool miscalibrates even perfectly calibrated individual

forecasts, and [6] study conditions for recovering calibration. Our objective is forecast accuracy (RPS), not calibration. The reliability diagrams in Figure 14 confirm all four methods are approximately calibrated on SPF data; the operative dimension of comparison is where the aggregate concentrates mass, not whether it is globally well-calibrated.

Scoring-rule-based combination. Forecast combination that minimises expected proper scoring-rule loss (see e.g. [16]) requires estimated accuracy weights for individual forecasters and is most natural when historical panel data are available. The Wasserstein barycenter requires no track record and imposes no model on the distribution of expert opinions, making it applicable to one-shot elicitations, new panels, and rare-event settings where accuracy estimation is infeasible.

Median aggregation of distribution functions. Hora et al. [11] propose aggregating expert distribution functions by taking the pointwise median of cumulative probabilities at each outcome value, studying both “vertical” (CDF median) and “horizontal” (quantile median) variants in the context of expert elicitation. Their vertical CDF median is operationally identical to the W_1 barycenter characterized in Proposition 2.5 below; our contribution is an optimal-transport foundation for their empirical recommendation together with the robustness analysis (Section 3), the ℓ^1 – ℓ^2 mismatch mechanism, and the adaptive blend that Hora et al. do not develop. In the optimal-transport literature, Carlier et al. [12] characterize the Wasserstein median for continuous distributions on $\mathbb{R}$, proving existence, uniqueness, robustness bounds, and PDE structure; their Section 4 establishes that the Wasserstein-1 median has CDF equal to the pointwise median of individual CDFs, parallel to our Proposition 2.5 in the discrete ordered setting. Our Proposition 2.5 is a discrete-space instance of Carlier et al.’s results, derived independently via a CDF-separability argument; we include it for self-containedness and to establish the notation used in our robustness analysis. The primary theoretical novelty of this paper is Theorem 2.2 (the ℓ^1 simplification for the indicator cost on unordered finite spaces), which has no analogue in the continuous OT literature. For the ordered case, the “horizontal” complement — the pointwise median

of quantile functions, or Vincentization — has been studied by Busetti [13] in the density-forecast context; the vertical (CDF-median) and horizontal (quantile-median) formulations generally differ and Carlier et al. discuss both.

Barycenter-based combination. For distributions on $\mathbb{R}$, the W_2 barycenter equals the quantile-mean of individual quantile functions [22]; [10] apply this to ensemble forecasting in climatology. Taylor [14] proposes depth-based median aggregates for CDF forecasts and applies them to ECB SPF inflation histograms, the same dataset we study in Section ??; we discuss the relationship in detail there. Our setting (finite categorical outcome spaces) enables a closed-form theory unavailable in the continuous case: the ℓ^1 simplification (Theorem 2.2) and the CDF-median formula (Proposition 2.5) provide an analytically tractable characterization that does not require a solver and yields sharp predictions about the regimes in which each method is preferred. Reweighting by past accuracy [4, 21, 8] is complementary: our approach and track-record reweighting could be combined in a weighted barycenter, but we study the equal-weight case to isolate the contribution of geometry.

This paper makes three contributions.

Theory. The primary theoretical contribution is Theorem 2.2: for finite unordered outcome spaces with the indicator ground cost, the Wasserstein objective simplifies exactly to a constrained ℓ^1 problem in the probability masses. This simplification — with no analogue in continuous OT theory — makes computation tractable and robustness analysis explicit. For ordered outcome spaces, Proposition 2.5 gives a fully closed-form barycenter: the distribution whose CDF is the component-wise median of the individual CDFs. This recovers the estimator of Hora et al. [11] via the optimal-transport route and is a discrete instance of Carlier et al. [12]; we present it for completeness and as the basis for the robustness analysis that follows. We prove nonexpansiveness of the CDF-median map under panel perturbations and a margin-based local constancy result (Corollary 2.8) that predicts when the barycenter dominates empirically. We prove a formal ℓ^1 – ℓ^2 mismatch result (Proposition 2.9): since

RPS equals the squared ℓ^2 CDF distance to the true outcome distribution plus a constant, the W_1 barycenter achieves lower expected RPS than the arithmetic mean if and only if it is closer to the truth in ℓ^2 . This gives regime conditions for each method’s dominance and organises the entire empirical evaluation: the barycenter wins when it is held near the truth by its local-constancy property (low-dispersion, consensus panels), while the arithmetic mean recovers ground when the panel is genuinely divided and tail outcomes materialize.

Empirical evaluation. We validate the framework across three independent datasets that differ along every relevant dimension: forecast format (binary vs. probability histogram), domain (geopolitical forecasting vs. macroeconomics), region (United States vs. Euro area), and institution (public tournament vs. professional central-bank survey). We benchmark against the three canonical alternatives: the arithmetic mean (AM), the logarithmic opinion pool (LP), and the symmetrically trimmed arithmetic mean (TM). (i) On 94 binary questions from the Good Judgment Project (GJP Year 1, ~ 900 forecasters each), the ℓ^1 barycenter reduces mean Brier score by 12.5% ($p < 0.001$); on 113 multi-category questions ($K \in \{3, 4, 5\}$, GJP Years 1–4), the indicator-cost ℓ^1 barycenter reduces mean Brier score by 8.7% ($p < 0.001$), with the reduction growing with K (3.1% for $K = 3$, 12.4% for $K = 4$, 17.2% for $K = 5$). (ii) On 76 quarters of U.S. Survey of Professional Forecasters core CPI histograms (2007–2025), the W_1 barycenter and the trimmed mean both achieve lower mean RPS than the AM (Wilcoxon $p = 0.005$; DM $p = 0.23$, the latter reflecting variance inflation from 2021–2022), while the LP has the worst RPS despite the highest P(correct bin); the barycenter’s advantage is concentrated in low-dispersion quarters (DM $p < 0.001$ in that regime). (iii) A replication on the ECB Survey of Professional Forecasters Euro-area HICP histograms confirms the dispersion-conditional pattern on an independent panel (DM $p = 0.006$ in low-dispersion quarters). The dispersion-conditional advantage therefore holds across all three datasets, two continents, and two forecast formats. A secondary opioid-survey application illustrates the unordered case with $K = 20$ outcomes [9].

Adaptive blending. We propose a dispersion-adaptive convex blend of the barycenter and the arithmetic mean as a distinct practical contribution not in Hora et al., Carlier et al., or Taylor [14]. The blend weight λ_t is estimated from an expanding historical window using only the panel’s past forecast record: no external data, no expert labeling, and no manually chosen tuning parameter. The blend is theoretically motivated: Corollary 2.8 identifies low cross-sectional dispersion as the regime in which the barycenter’s local-constancy property activates, and the estimated λ_t tracks this regime boundary automatically without a binary dispersion switch. Empirically, the adaptive blend achieves the lowest out-of-sample RPS of any method evaluated (0.404), with a statistically significant improvement over the arithmetic mean (4.3%, $p < 0.001$); its margin over the trimmed mean is smaller (1.1%) and not individually significant. The mixture weight converges to a stable value within the first 20 quarters of the expanding window, indicating the blend is usable early in a panel’s life. We view the blend as a contribution independent of the barycenter: it operationalizes the theoretical regime structure as a data-driven estimator without manual tuning, achieving gains over the arithmetic mean that neither the pure barycenter nor the trimmed mean fully captures.

The remainder of the paper is organized as follows. Section 2 states the two main theoretical results: the ℓ^1 simplification theorem for unordered spaces and the CDF-median characterization for ordered spaces. Section 3 formalizes robustness via the empirical influence function. Section 4 reports simulation evidence. Section 5 presents a scored binary evaluation using the Good Judgment Project. Section 6 presents the U.S. SPF application: single-quarter illustration, 76-quarter historical backtest, dispersion-adaptive blend, and ECB SPF replication. Section 7 concludes.

2 Wasserstein Barycenters as Aggregate Measures

Let $\Omega = \{1, \dots, K\}$ be a finite set of outcomes. Write $P \in \mathbb{R}^{N \times K}$ for the matrix of expert probability vectors, with row $P_j \in \Delta_{K-1}$ representing expert j 's distribution, and let D be the $K \times K$ matrix of ground distances between outcomes. Because Δ_{K-1} is compact and the Wasserstein objective is continuous, a barycenter exists for any ground cost [22]. The main result of this section is that when D is the indicator distance, the barycenter objective simplifies to an ℓ^1 problem in the probability masses (Theorem 2.2), giving a computationally tractable and geometrically transparent characterization. The ordered case ($d(x, y) = |x - y|$) yields a further closed-form result (Proposition 2.5).

2.1 Definition of the Wasserstein Barycenter

The following definition can be found in [24]. Let (Ω, d) be a metric space for which every measure is a Radon measure and let $\mathcal{P}(\omega)$ be the set of measures on Ω with finite p^{th} moment for some element in Ω .

Definition 1. A Wasserstein barycenter of N measures $\{P_1, \dots, P_N\}$ is a minimizer of

$$f(\Psi) = \frac{1}{N} \sum_{i=1}^N W_p^p(\Psi, P_i).$$

over the set of all measures over the state space Ω where W_p is the Wasserstein distance given by

$$W_p = \left(\inf_{\pi \in \Gamma(\mu, \nu)} \int d(x, y)^p d\pi(x, y) \right)^{1/p}.$$

The authors of [23] present results for the support of discrete Wasserstein barycenter for finitely supported measures. That is, each measure places mass on a finite number of points in Euclidean space. The authors show the set of support points is not contained in the union of the support of the individual measures. Hence discrete Wasserstein barycenters

in Euclidean space are not bounded by the component probability measures. But, we are concerned with the case such that $\Omega = \{1, \dots, K\}$. The definition becomes any Ψ that minimizes

$$\sum_{j=1}^N \left(\min_{\pi \in \Gamma(p_j, \Psi)} \sum_{x=1}^K \sum_{y=1}^K d(x, y)^p \pi(x, y) \right).$$

We consider two ground costs: the indicator distance for unordered outcome spaces (Section 2.2) and the absolute-value distance $d(x, y) = |x - y|$ for ordered spaces.

Remark 1 (Barycenter, median, and specialization to $p = 1$). *Throughout this paper we specialize to $p = 1$. For the indicator cost, $W_{\text{ind}}^2 = W_{\text{ind}}$ (the cost takes values in $\{0, 1\}$), so the $p = 1$ and $p = 2$ barycenter objectives coincide up to a positive constant. For the ordered case (Section ??), the objective $\sum_j W_1(\Psi, P_j)$ minimizes a sum of distances rather than squared distances; this is sometimes called the Wasserstein median rather than barycenter, following the nomenclature of Carlier et al. [12]. We retain “barycenter” for consistency with the forecast-combination literature [22, 10] and note that the distinction does not affect any result in this paper.*

2.2 Unordered Finite Outcome Space

When the distance function is given by

$$d(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

225 choose matrices $\Pi^1, \dots, \Pi^N$ to minimize the objective function

$$\begin{aligned} f(\Pi) &= \sum_{l=1}^N \left(\sum_{k=1}^K \sum_{k'=1}^K d(k, k')^2 \Pi_{k, k'}^l \right) \\ &= \sum_{l=1}^N \sum_{k=1}^K \sum_{k' \neq k} \Pi_{k, k'}^l \end{aligned}$$

226 subject to the feasibility conditions for $j = 1, \dots, N$ and $l, m = 1, \dots, N$

$$\begin{aligned} \sum_j \Pi_{k, j}^l &= p_l(k) \\ \sum_j \Pi_{j, k}^l &= \sum_j \Pi_{j, k}^m. \end{aligned}$$

227 The key identity needed for the main theorem is the following.

228 **Lemma 2.1** (Indicator-cost identity). *For $p, \Psi \in \Delta_{K-1}$,*

$$W_{\text{ind}}(p, \Psi) = 1 - \sum_k \min\{p(k), \Psi(k)\} = \frac{1}{2} \|p - \Psi\|_1.$$

229 *Proof.* The transport cost under the indicator distance is $1 - \sum_k \Pi_{kk}$ for any coupling $\Pi \in$
 230 $\Gamma(p, \Psi)$, minimized by maximising $\sum_k \Pi_{kk}$. Marginal constraints give $\Pi_{kk} \leq \min\{p(k), \Psi(k)\}$,
 231 and the plan $\Pi_{kk}^* = \min\{p(k), \Psi(k)\}$ is feasible (residual row and column masses are nonneg-
 232 ative and sum to the same total, so an off-diagonal assignment exists). Thus $W_{\text{ind}}(p, \Psi) =$
 233 $1 - \sum_k \min\{p(k), \Psi(k)\}$. The second equality uses $a - \min(a, b) = \max(a - b, 0)$ and
 234 $\sum_k (p(k) - \Psi(k)) = 0$ to give $\sum_k \max\{p(k) - \Psi(k), 0\} = \frac{1}{2} \|p - \Psi\|_1$. $\square$

235 The analogous property for barycenter solutions (that every optimal plan simultaneously
 236 satisfies $\Pi_{kk}^l = \min\{p_l(k), \Psi(k)\}$) is established in Appendix C.

237 **Theorem 2.2** (ℓ^1 Simplification). *Let $\{p_1, \dots, p_N\}$ be probability mass functions on $\{1, \dots, K\}$*

238 and let the ground cost be the indicator distance. Then the Wasserstein barycenter Ψ^* solves

$$\min_{\Psi \in \Delta_{K-1}} \sum_{l=1}^N \sum_{k=1}^K |p_l(k) - \Psi(k)|, \quad (2.1)$$

239 the ℓ^1 sum of componentwise deviations from the experts. In particular, a Wasserstein
 240 barycenter exists and the Wasserstein objective is equivalent to an ℓ^1 regression of Ψ onto
 241 the expert opinions.

242 *Proof.* Fix any candidate $\Psi \in \Delta_{K-1}$. By Lemma 2.1, the optimal transport cost $\text{OT}(\Psi, p_l)$
 243 between Ψ and each p_l under the indicator distance equals

$$\text{OT}(\Psi, p_l) = 1 - \sum_k \min\{p_l(k), \Psi(k)\} = \frac{1}{2} \sum_k |p_l(k) - \Psi(k)|,$$

244 using the identity $a - \min(a, b) = \max(a - b, 0)$ and the fact that $\sum_k (p_l(k) - \Psi(k)) = 0$.
 245 Summing over l gives the objective (2.1) up to a positive constant. The function $\Psi \mapsto$
 246 $\sum_{l,k} |p_l(k) - \Psi(k)|$ is continuous on the compact simplex, so the minimum is attained. $\square$

247 **Remark 2.** The identity $\text{OT}(p, \Psi) = \frac{1}{2} \|p - \Psi\|_1$ is the standard equivalence between the
 248 Wasserstein-1 distance under the indicator cost and the total variation distance; see [19] for a
 249 survey. The theorem reduces the barycenter problem to ℓ^1 median estimation on the simplex.
 250 The unconstrained minimizer assigns each state the coordinatewise median of the expert
 251 opinions (subject to the normalization constraint $\sum_k \Psi(k) = 1$), and the simplex projection
 252 shifts this toward the feasible median. The arithmetic mean generally does not satisfy this
 253 condition and therefore does not generally coincide with the Wasserstein barycenter.

254 **Corollary 2.3** (Boundedness). Every Wasserstein barycenter Ψ^* under the indicator cost
 255 satisfies

$$\min_j p_j(k) \leq \Psi^*(k) \leq \max_j p_j(k) \quad \text{for each } k = 1, \dots, K.$$

256 In particular, Ψ^* assigns zero probability to any outcome that no expert assigns positive
 257 probability.

258 *Proof.* By the Full-Diagonal Criterion (Proposition C.1), every optimal transport plan sat-
 259 isfies $\Pi_{kk}^l = \min\{p_l(k), \Psi^*(k)\}$ for all l and k . Suppose $\Psi^*(k) > \max_j p_j(k)$ for some k .
 260 Then $\min\{p_l(k), \Psi^*(k)\} = p_l(k)$ for every l , so the diagonal entries in column k account
 261 for only $\sum_l p_l(k) < N\Psi^*(k)$ units of the column mass $N\Psi^*(k)$. The remaining mass
 262 $N\Psi^*(k) - \sum_l p_l(k) > 0$ must come from off-diagonal entries in column k , each incurring
 263 transport cost 1. Reducing $\Psi^*(k)$ by any $\varepsilon > 0$ small enough (and increasing some other
 264 coordinate to maintain the simplex constraint) would eliminate some of this off-diagonal
 265 mass, strictly decreasing the objective. This contradicts optimality, so $\Psi^*(k) \leq \max_j p_j(k)$.
 266 The lower bound $\Psi^*(k) \geq \min_j p_j(k)$ follows symmetrically by considering row masses. $\square$

267 **Remark 3** (Unanimity). *If all experts agree ($p_1 = p_2 = \dots = p_N = p$), then $\Psi^* = p$. This*
 268 *follows immediately from Corollary 2.3: each bound collapses to $p(k)$, so $\Psi^*(k) = p(k)$ for*
 269 *every k .*

270 **Corollary 2.4** (Binary case). *For $K = 2$, the Wasserstein barycenter under the indicator*
 271 *cost is*

$$\Psi^*(1) = \text{median}\{p_1(1), \dots, p_N(1)\}, \quad \Psi^*(2) = 1 - \Psi^*(1).$$

272 *Proof.* With $K = 2$ the simplex constraint $\Psi(1) + \Psi(2) = 1$ reduces the optimisation to a
 273 single free variable. The objective $\sum_l (|p_l(1) - \Psi(1)| + |p_l(2) - \Psi(2)|) = 2 \sum_l |p_l(1) - \Psi(1)|$
 274 is minimised by the median of $\{p_l(1)\}$, which automatically lies in $[0, 1]$ and hence satisfies
 275 the simplex constraint. $\square$

276 **Remark 4** (Breakdown point). *The finite-sample replacement breakdown point of the binary*
 277 *barycenter is $(\lfloor N/2 \rfloor + 1)/N \approx 50\%$, matching the classical sample median: one must corrupt*
 278 *strictly more than half the panel to drive $\Psi^*(1)$ to an arbitrary value. Any minority of*
 279 *corrupted opinions, however extreme, leaves $\Psi^*(1)$ unchanged provided the corrupted values*
 280 *do not cross the median rank. This is a precise statement of robustness: the barycenter is*
 281 *locally constant (Corollary 2.8 gives the exact margin) and globally stable under minority*
 282 *corruption. By contrast, the arithmetic mean has breakdown point $1/N$: a single extreme*

283 *opinion shifts the aggregate, and its influence grows without bound as $P_0 \rightarrow \delta_k$.*

284 **2.3 Discrete Barycenters for Ordered Spaces**

285 When the outcome space $\Omega = \{1, \dots, K\}$ carries a natural ordering, the indicator distance
 286 used in the preceding section discards structural information. Setting $d(x, y) = |x - y|$ makes
 287 (Ω, d) a metric space that respects the ordering and gives rise to a distinct class of Wasserstein
 288 barycenters. Let $F_j(k) = \sum_{x \leq k} P_j(x)$ denote the cumulative distribution function (CDF) of
 289 P_j and let $F_j^{-1}(t) = \inf \{k : F_j(k) \geq t\}$ denote its quantile function.

290 **Wasserstein-1 Barycenters**

291 For $p = 1$, the Wasserstein distance between P and Q on Ω reduces to the L^1 distance
 292 between their CDFs [18, 19]:

$$W_1(P, Q) = \sum_{k=1}^{K-1} |F_P(k) - F_Q(k)|.$$

293 The barycenter minimizes $\sum_{j=1}^N W_1(\Psi, P_j)$ over probability mass functions Ψ on Ω . Substi-
 294 tuting the CDF representation and exchanging the order of summation gives

$$\sum_{j=1}^N W_1(\Psi, P_j) = \sum_{k=1}^{K-1} \sum_{j=1}^N |F_\Psi(k) - F_j(k)|,$$

295 which separates across k . Each term is independently minimized by the sample median of
 296 $\{F_j(k)\}_{j=1}^N$.

297 **Proposition 2.5.** *Let $\{P_1, \dots, P_N\}$ be probability mass functions on $\{1, \dots, K\}$ and let W_1
 298 be defined with $d(x, y) = |x - y|$. A Wasserstein-1 barycenter has CDF*

$$F_\Psi(k) = \text{med} \{F_1(k), \dots, F_N(k)\}, \quad k = 1, \dots, K - 1.$$

Remark 5. *This result is a discrete-space instance of Carlier et al. [12], whose Proposition 4.1 establishes the analogous characterization for continuous distributions on $\mathbb{R}$. The same estimator is advocated on empirical grounds by Hora et al. [11] under the name “vertical median aggregation.” We include the proof for self-containedness. Uniqueness of the W_1 barycenter on a finite ordered grid follows from the strict-convexity argument of Carlier et al.; in our discrete setting with N odd it reduces to the observation that the componentwise median of the CDFs is uniquely determined whenever no two CDFs coincide at the median rank.*

Proof. Since the objective separates across k , the unconstrained minimizer at each k is the sample median of $\{F_j(k)\}_{j=1}^N$. It remains to verify that the vector of component-wise medians constitutes a valid CDF, i.e., it is nondecreasing in k . Since each F_j is nondecreasing, $F_j(k) \leq F_j(k+1)$ for every j . If $x_j \leq y_j$ for all j , then $\text{med}\{x_j\} \leq \text{med}\{y_j\}$; hence $F_\Psi(k) \leq F_\Psi(k+1)$ for all k . The boundary conditions $F_\Psi(0) = 0$ and $F_\Psi(K) = 1$ hold since $F_j(0) = 0$ and $F_j(K) = 1$ for every j . Thus F_Ψ is a valid CDF and $\Psi(k) = F_\Psi(k) - F_\Psi(k-1) \geq 0$ defines a probability mass function on Ω . $\square$

The W_1 barycenter has a natural interpretation: it is the distribution whose CDF lies at the “center” of the collection in the L^1 sense.

Corollary 2.6 (CDF Boundedness). *The W_1 barycenter Ψ^* satisfies*

$$\min_j F_j(k) \leq F_{\Psi^*}(k) \leq \max_j F_j(k) \quad \text{for each } k = 1, \dots, K-1.$$

In particular, Ψ^ never assigns cumulative probability outside the range spanned by the panel.*

Proof. Immediate from Proposition 2.5: the component-wise median of $\{F_j(k)\}$ lies in $[\min_j F_j(k), \max_j F_j(k)]$. $\square$

Robustness of the Ordered Barycenter

The CDF-median structure of the W_1 barycenter implies two stronger robustness results than the perturbation bound of Corollary 10.

Proposition 2.7 (CDF Nonexpansiveness). *Let F^* and $\tilde{F}^*$ denote the W_1 barycenter CDFs associated with panels $\{F_1, \dots, F_N\}$ and $\{\tilde{F}_1, \dots, \tilde{F}_N\}$. Then*

$$\|F^* - \tilde{F}^*\|_\infty \leq \max_{1 \leq j \leq N} \|F_j - \tilde{F}_j\|_\infty.$$

Proof. By Proposition 2.5, $F^*(k) = \text{med}\{F_j(k)\}$ and $\tilde{F}^*(k) = \text{med}\{\tilde{F}_j(k)\}$. The sample median is 1-Lipschitz in the ℓ^∞ sense: for any real vectors $x, y \in \mathbb{R}^N$, $|\text{med}(x) - \text{med}(y)| \leq \|x - y\|_\infty$. Applying this coordinatewise gives $|F^*(k) - \tilde{F}^*(k)| \leq \max_j |F_j(k) - \tilde{F}_j(k)| \leq \max_j \|F_j - \tilde{F}_j\|_\infty$. Taking the supremum over k proves the claim. $\square$

Remark 6. *Proposition 2.7 contrasts sharply with the arithmetic mean, for which $\|\bar{F} - \tilde{\bar{F}}\|_\infty \leq \frac{1}{N} \sum_j \|F_j - \tilde{F}_j\|_\infty$. The AM bound involves the average CDF perturbation, so a single large outlier contributes $1/N$ of its full shift. The barycenter bound involves the maximum perturbation; one very large shift dominates. This is median-like behavior — but the key practical implication runs in the opposite direction of a naive reading: the barycenter’s sup-norm bound is no worse than the largest single-panel perturbation, whereas the AM’s bound can be far smaller when outliers are isolated. What the bound does guarantee is that the barycenter cannot be destabilized by many small perturbations all in the same direction.*

Corollary 2.8 (Margin-Based Local Constancy). *Assume N is odd. For each $k = 1, \dots, K - 1$, define γ_k as the gap between the median CDF value and its nearest neighboring order statistic,*

$$\gamma_k = \min\{F_{(m)}(k) - F_{(m-1)}(k), F_{(m+1)}(k) - F_{(m)}(k)\},$$

where $F_{(1)}(k) \leq \dots \leq F_{(N)}(k)$ are the order statistics of $\{F_j(k)\}$ and $m = (N + 1)/2$. If

$$\max_{j,k} |\tilde{F}_j(k) - F_j(k)| < \min_k \gamma_k,$$

then $\tilde{F}^* = F^*$: the W_1 barycenter is completely unchanged by the perturbation.

Proof. Under the stated condition, the perturbation is strictly smaller than the median gap at every coordinate. Therefore the rank ordering of $\{\tilde{F}_j(k)\}$ is identical to that of $\{F_j(k)\}$ at each k , so the median order statistic is the same element. Hence $\tilde{F}^*(k) = F^*(k)$ for all k . $\square$

Remark 7. Corollary 2.8 formalizes why the barycenter is robust in consensus regimes: when the panel CDFs are tightly clustered (large γ_k), the aggregate is completely immune to perturbations smaller than the cluster gap. The arithmetic mean has no analogous immunity. Any perturbation, however small, shifts the mean CDF by exactly $1/N$ of that amount.

Connection to the Ranked Probability Score

The empirical evaluation in Section 6.3 uses the Ranked Probability Score (RPS) as its primary criterion. The following proposition formalizes the relationship between the W_1 barycenter, RPS, and the dispersion-conditional regime structure observed in the data.

Proposition 2.9 (ℓ^1 - ℓ^2 Mismatch). Let $F_{\text{true}}(k) = \Pr(b \leq k)$ denote the CDF of the true outcome distribution on $\{1, \dots, K\}$. For any forecast CDF F ,

$$\mathbb{E}_b[\text{RPS}(F, b)] = \|F - F_{\text{true}}\|_2^2 + c(F_{\text{true}}), \quad (2.2)$$

where $\|F - G\|_2^2 = \sum_{k=1}^{K-1} (F(k) - G(k))^2$ and $c(F_{\text{true}}) = \sum_{k=1}^{K-1} F_{\text{true}}(k)(1 - F_{\text{true}}(k))$ is a constant independent of F . Consequently:

(i) The W_1 barycenter F^* achieves lower expected RPS than the arithmetic mean $\bar{F}$ if and only if $\|F^* - F_{\text{true}}\|_2 < \|\bar{F} - F_{\text{true}}\|_2$.

(ii) If $F_{\text{true}} = F^*$ (the true outcome distribution equals the panel CDF-median), then

$$\mathbb{E}[\text{RPS}(F^*)] \leq \mathbb{E}[\text{RPS}(\bar{F})], \text{ with strict inequality unless } \bar{F} = F^*.$$

(iii) If $F_{\text{true}} = \bar{F}$ (the true outcome distribution equals the panel mean CDF), then $\mathbb{E}[\text{RPS}(\bar{F})] \leq$

$$\mathbb{E}[\text{RPS}(F^*)], \text{ with strict inequality unless } \bar{F} = F^*.$$

Proof. Since $\text{RPS}(F, b) = \sum_{k=1}^{K-1} (F(k) - \mathbf{1}_{b \leq k})^2$, taking expectation over b :

$$\mathbb{E}_b[\text{RPS}(F, b)] = \sum_{k=1}^{K-1} \mathbb{E}_b[(F(k) - \mathbf{1}_{b \leq k})^2] = \sum_{k=1}^{K-1} [(F(k) - F_{\text{true}}(k))^2 + F_{\text{true}}(k)(1 - F_{\text{true}}(k))],$$

where the second equality uses $\mathbb{E}_b[\mathbf{1}_{b \leq k}] = F_{\text{true}}(k)$ and $\mathbb{E}_b[\mathbf{1}_{b \leq k}^2] = F_{\text{true}}(k)$. This establishes (2.2). Parts (i)–(iii) follow immediately: the constant $c(F_{\text{true}})$ cancels in comparisons, so expected-RPS ordering reduces entirely to ℓ^2 -distance ordering relative to F_{true} . $\square$

Remark 8. Equation (2.2) is the standard Bregman decomposition for proper scoring rules; the RPS is strictly proper with Bayes act F_{true} . What Proposition 2.9 adds is the regime characterization: since the W_1 barycenter minimizes an ℓ^1 objective in CDF space (Proposition 2.5) while expected RPS is an ℓ^2 criterion, the optimal forecast for each objective is generically different. Part (ii) is the consensus-regime result: when a minority of outliers displace $\bar{F}$ from F_{true} while the local-constancy property (Corollary 2.8) holds F^* near F_{true} , the barycenter wins. Part (iii) is the high-dispersion result: when the panel is genuinely divided and no systematic outlier pattern is present, the AM’s equal-weight averaging may place $\bar{F}$ closer to F_{true} than F^* . Corollary 2.8 identifies the margin γ_k as the structural quantity governing which regime applies: large γ_k (tight consensus) activates the local-constancy that makes F^* stable near F_{true} ; small γ_k (dispersed panel) means both aggregates move with panel fluctuations and neither has a structural advantage.

3 Robustness of Aggregating Measures

Both the arithmetic mean and the Wasserstein barycenter are bounded aggregating measures, as established in the preceding sections. They differ, however, in their sensitivity to extreme or outlying expert opinions. This section formalizes that difference via the empirical influence function and illustrates it through simulation.

3.1 Empirical Influence Function

The empirical influence function (EIF) quantifies the effect of introducing a single additional expert P_0 into an existing collection $\mathbf{P}_N = \{P_1, \dots, P_N\}$. For an aggregating measure T , the EIF is

$$\text{EIF}(P_0; T, \mathbf{P}_N) = (N + 1) [T(\mathbf{P}_N \cup \{P_0\}) - T(\mathbf{P}_N)],$$

rescaled so that it approximates the derivative of T with respect to contamination fraction $\varepsilon = 1/(N + 1)$. The gross error sensitivity $\text{GES}(T) = \sup_{P_0} \|\text{EIF}(P_0; T, \mathbf{P}_N)\|_\infty$ summarizes the worst-case influence of a single outlier.

Proposition 3.1 (EIF of the Arithmetic Mean). *Let $\bar{P}_N = \frac{1}{N} \sum_{j=1}^N P_j$ be the arithmetic mean of $\mathbf{P}_N$. Then*

$$\text{EIF}_k(P_0; \bar{P}, \mathbf{P}_N) = P_0(k) - \bar{P}_N(k)$$

for each state $k \in \Omega$.

Proof. Adding P_0 to $\mathbf{P}_N$ gives $\bar{P}_{N+1}(k) = \frac{N\bar{P}_N(k) + P_0(k)}{N+1}$, so

$$(N + 1) [\bar{P}_{N+1}(k) - \bar{P}_N(k)] = P_0(k) - \bar{P}_N(k). \quad \square$$

Remark 9. *The EIF of the arithmetic mean is linear in P_0 and independent of N . Every*

397 *opinion, regardless of how extreme, exerts unit influence on the aggregate. The gross error*
 398 *sensitivity is $\text{GES}(\bar{P}) = \sup_{P_0} \|P_0 - \bar{P}_N\|_\infty \leq 1 - \min_k \bar{P}_N(k)$, which approaches 1 as $P_0 \rightarrow \delta_k$*
 399 *for any k .*

400 The exact EIF of the Wasserstein barycenter does not admit a closed form in general,
 401 because the barycenter is defined as the solution to a nonlinear variational problem whose
 402 geometry changes when P_0 is added. A useful bound is available, however, from the bound-
 403 edness result of Corollary 2.3.

404 **Remark 10** (Range bound on barycenter shift). *Let $\hat{\Psi}_N$ and $\hat{\Psi}_{N+1}$ denote the Wasserstein*
 405 *barycenters of $\mathbf{P}_N$ and $\mathbf{P}_N \cup \{P_0\}$ under the indicator cost. By Corollary 2.3, both lie within*
 406 *the cross-sectional range of their respective panels, so*

$$\left| \hat{\Psi}_{N+1}(k) - \hat{\Psi}_N(k) \right| \leq \max_{j \leq N} P_j(k) \vee P_0(k) - \min_{j \leq N} P_j(k) \wedge P_0(k)$$

407 *for each k . This is a direct consequence of the boundedness result and not an independent*
 408 *robustness claim: the bound equals the full cross-sectional range and can be close to 1 when the*
 409 *panel is heterogeneous. The operative robustness characterizations are the EIF comparison*
 410 *above, the CDF nonexpansiveness (Proposition 2.7), and the margin-based local constancy*
 411 *(Corollary 2.8).*

412 The two results reveal a structural asymmetry. The EIF of the arithmetic mean equals
 413 $P_0 - \bar{P}_N$ exactly, independent of N : every opinion, however extreme, shifts the aggregate
 414 by a fixed amount. For the barycenter, the breakdown point analysis (Remark following
 415 Corollary 2.4) shows that a single outlier cannot move the aggregate at all provided it does
 416 not cross the median rank — a qualitatively different robustness guarantee than the AM’s
 417 linear sensitivity. Stronger results for the ordered case are in Section 2.

3.2 Simulation Evidence

To illustrate the practical difference, we simulate $R = 200$ collections of $N = 11$ experts. Ten experts are drawn from a uniform Dirichlet distribution over $K = 6$ states. One expert is drawn from a concentrated Dirichlet placing substantial mass on a single state (the “outlier”). For each collection, both the arithmetic mean and the Wasserstein barycenter are computed.

Figure 1 displays the distribution of the probability assigned to the outlier’s preferred state (S_6) across the R replicates. The arithmetic mean is substantially more inflated: in 90% of replicates the AM assigns more probability to S_6 than the barycenter does, consistent with its equal-weight treatment of the outlier. Figure 2 shows the variance of each aggregate across replicates by state. The barycenter has lower variance on the outlier state (0.0039 vs. 0.0016 for the AM on S_6) but higher overall variance, confirming the classical robustness–efficiency tradeoff.

Figure 3 displays a single representative draw: five regular experts, one outlier, and the two resulting aggregates. The arithmetic mean is visibly pulled toward the outlier’s distribution; the barycenter remains closer to the consensus of the regular experts.

Figure 4 varies the number of outlying experts from zero to three. The arithmetic mean’s assignment to S_6 grows approximately linearly with outlier count. The barycenter grows more slowly, reflecting its distributional resistance.

Remark 11. *The higher variance of the barycenter across replicates is expected. When expert opinions are drawn from a well-specified population, the arithmetic mean is the minimum-variance unbiased estimator of the population mean. The barycenter trades efficiency for robustness, analogous to the median relative to the mean in location estimation. The practical choice between the two depends on the degree of belief in the homogeneity of the expert pool.*

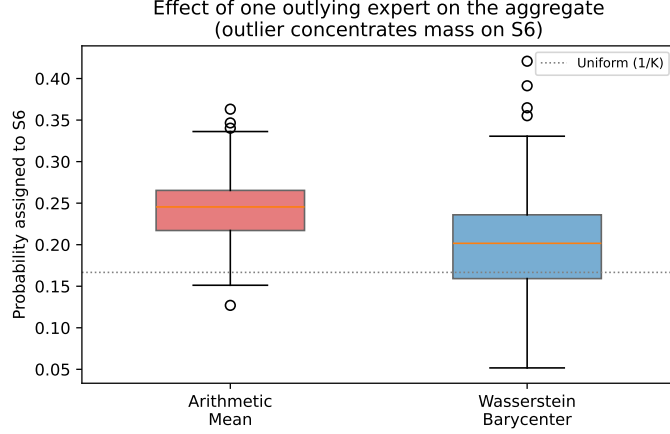


Figure 1: Distribution of the probability assigned to the outlier’s preferred state (S_6) across $R = 200$ simulation replicates. The arithmetic mean (red) is systematically more inflated than the Wasserstein barycenter (blue).

4 Simulations

The robustness analysis of the preceding section held the structure of contamination fixed: one outlying expert drawn from a concentrated Dirichlet, $N - 1$ regular experts drawn from a uniform Dirichlet. This section broadens the comparison through a systematic simulation study examining (1) efficiency under a correctly specified population model and (2) robustness under varying degrees of misspecification.

4.1 Setup

All simulations use $K = 6$ states. Expert opinions are drawn from one of two Dirichlet populations:

- **Regular experts:** $P_j \sim \text{Dir}(\mathbf{1}_K)$, the uniform Dirichlet placing equal prior weight on all states.
- **Outlying experts:** $P_j \sim \text{Dir}(\alpha_{\text{out}})$ with $\alpha_{\text{out}} = (1, 1, 1, 1, 1.5, 10)^{\odot 4}$, a highly concentrated distribution placing substantial mass on state S_6 .

For each simulation, N_{reg} regular and N_{out} outlying opinions are drawn; the arithmetic mean and Wasserstein barycenter (with uniform weights and indicator distance) are computed;

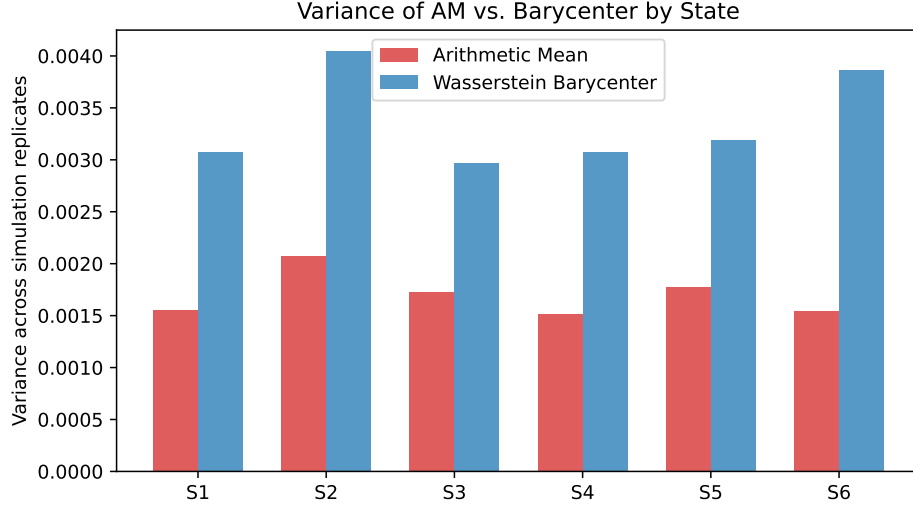


Figure 2: Variance of the arithmetic mean and Wasserstein barycenter by state across $R = 200$ replicates. The barycenter has lower variance on the outlier state (S_6) but higher variance on regular states, illustrating the robustness–efficiency tradeoff.

and the results are summarized across $R = 200$ replicates. Performance is measured by (i) mean squared error relative to the population mean of the regular experts, (ii) variance across replicates, and (iii) probability assigned to the outlier’s preferred state S_6 .

4.2 Efficiency Under the Null

When all experts are drawn from the same regular population ($N_{\text{out}} = 0$), the arithmetic mean is the minimum-variance unbiased estimator of the population probability vector. Table 1 compares the mean squared error of the two aggregates relative to the true population mean $\mu = \mathbf{1}_K/K$ for varying N .

N	AM MSE	BC MSE	BC/AM ratio
5	0.0225	0.0317	1.41
10	0.0124	0.0198	1.59
20	0.0063	0.0118	1.86

Table 1: Mean squared error (summed over states) of the arithmetic mean and Wasserstein barycenter relative to the true population mean $\mu = \mathbf{1}_K/K$, averaged over $R = 200$ replicates with no outlying experts. The AM is more efficient in all cases; the ratio grows with N .

Both estimators are consistent: MSE decreases at rate $O(1/N)$. The barycenter’s ef-

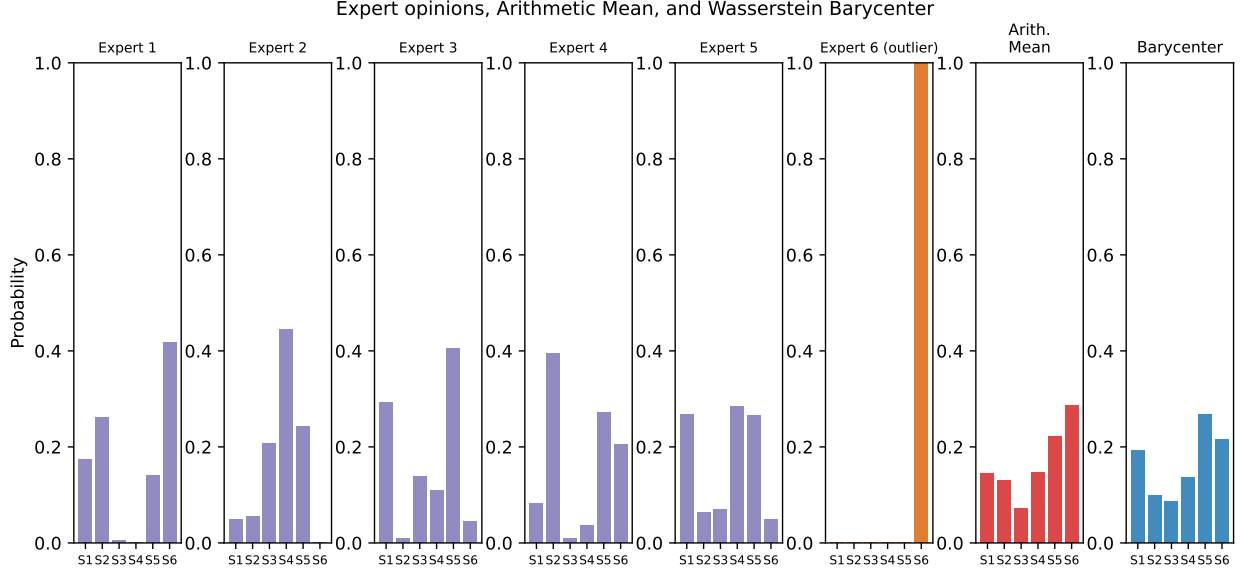


Figure 3: A single simulation draw: five regular experts (purple), one outlier (orange), arithmetic mean (red), and Wasserstein barycenter (blue). The arithmetic mean is visibly pulled toward the outlier’s distribution.

465 efficiency loss relative to the AM grows with N , reaching a ratio of 1.86 at $N = 20$. This
466 reflects the L^1 vs. L^2 character of the two estimators: as more data arrives, the AM concen-
467 trates its distribution more tightly, while the barycenter’s Wasserstein objective introduces
468 a systematic dispersion penalty.

469 4.3 Robustness Under Contamination

470 Figure 4 (Section 3) already showed that the barycenter resists outliers more effectively as
471 their count increases from zero to three. Here we quantify the tradeoff more precisely. For
472 $N_{\text{reg}} = 10$ and $N_{\text{out}} \in \{0, 1, 2, 3\}$, Table 2 reports the mean and standard deviation of the
473 probability assigned to S_6 under each aggregate.

474 The arithmetic mean’s assignment to S_6 increases by $\approx 1/N_{\text{tot}}$ per additional outlier,
475 consistent with the EIF derived in Section 3. The barycenter’s increase is approximately
476 $1.7\times$ smaller, and its standard deviation rises modestly as expert opinions become more
477 heterogeneous, while the AM standard deviation falls (the outlier deterministically anchors
478 the mean toward S_6).

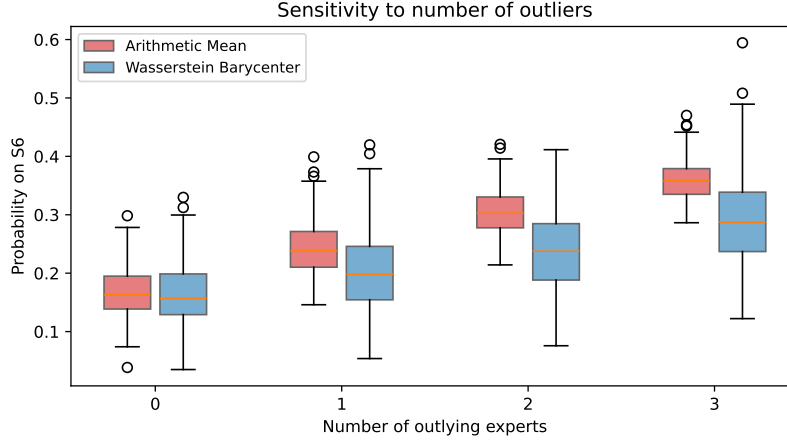


Figure 4: Sensitivity to outlier count. As the number of outlying experts increases from zero to three, the arithmetic mean’s assignment to S_6 grows more rapidly than the barycenter’s.

N_{out}	AM mean (S_6)	AM sd	BC mean (S_6)	BC sd
0	0.165	0.044	0.166	0.057
1	0.243	0.041	0.202	0.061
2	0.309	0.040	0.245	0.075
3	0.357	0.033	0.281	0.074

Table 2: Mean and standard deviation of the probability assigned to state S_6 as the number of outlying experts increases. The AM mean grows by approximately 0.064 per additional outlier; the BC mean grows by approximately 0.038.

4.4 Sensitivity to Population Model

A natural question is how performance depends on the degree of heterogeneity in the regular expert population. We vary the Dirichlet concentration parameter α_0 over a log-spaced grid from 0.25 to 20 (with $\alpha = \alpha_0 \mathbf{1}_K$), where smaller values produce more dispersed expert opinions. Figure 5 shows that the MSE ratio decreases monotonically as α_0 increases, falling from 2.45 at $\alpha_0 = 0.25$ to 1.35 at $\alpha_0 = 20$. When experts are highly dispersed (small α_0), the barycenter’s ℓ^1 objective assigns heavier weight to large deviations, amplifying its efficiency loss relative to the ℓ^2 -optimal AM. As the population concentrates, expert opinions converge and the two aggregates become more similar.

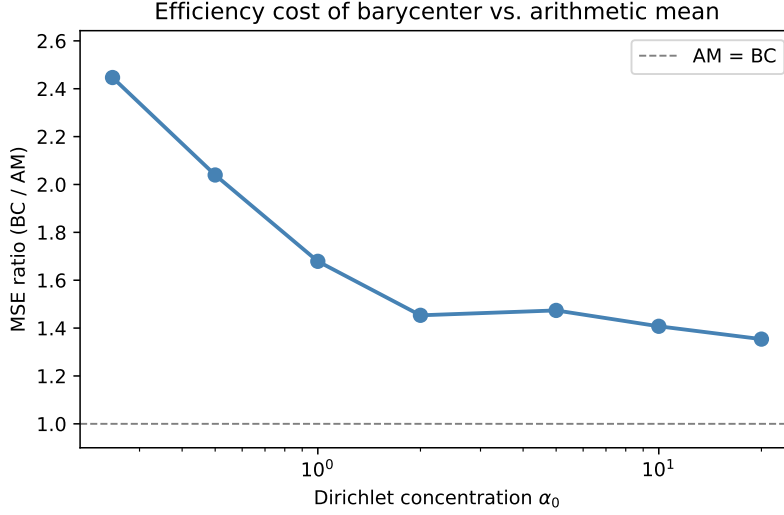


Figure 5: MSE ratio (BC/AM) as a function of the Dirichlet concentration α_0 , with $N = 10$ regular experts and no outliers. The ratio is stable across population models.

5 Good Judgment Project

We use data from the Good Judgment Project (GJP), a geopolitical forecasting tournament conducted under the IARPA ACE program from 2011 to 2015 [2]. The GJP dataset contains both binary questions (answered with a single $P(\text{Yes})$) and multi-category questions (answered with a probability vector over $K \geq 3$ unordered outcomes). This structure allows us to evaluate both the binary barycenter (Corollary 2.4) and the full indicator-cost ℓ^1 barycenter (Theorem 2.2) in a common scored setting.

5.1 Binary Questions

For binary outcomes, Corollary 2.4 gives that the indicator-cost ℓ^1 barycenter reduces to the component-wise median of the individual probability vectors. Each forecaster’s distribution over $\{\text{Yes}, \text{No}\}$ is characterized by a single number $p_j \in [0, 1]$; the barycenter selects $p^* = \text{median}_j(p_j)$ while the arithmetic mean selects $\bar{p} = \frac{1}{N} \sum_j p_j$. The dispersion-robustness intuition is direct: if a minority of forecasters assigns extreme probabilities near 0 or 1, the median is unaffected while the mean is pulled toward them.

Data and protocol. We use individual survey forecasts from Year 1 of the GJP (2011–2012), publicly available through Harvard Dataverse [3]. For each of the 94 binary questions with at least 5 participating forecasters (mean panel size: 895), we take each forecaster’s last submitted forecast before question closure and compute the arithmetic mean and median of the individual $P(\text{Yes})$ values. We score each aggregate against the binary resolution using the Brier score, $\text{BS} = (p - \mathbf{1}_{\text{Yes}})^2$.

Results. The median (barycenter) achieves a lower Brier score than the arithmetic mean on 85.1% of questions, with mean Brier scores of 0.091 (BC) versus 0.104 (AM), a 12.5% improvement ($p < 0.001$, Wilcoxon). The advantage is present across all dispersion terciles and is largest when the panel exhibits low disagreement (Table 3 and Figure 6).

Dispersion group	AM Brier	BC Brier	BC improvement
Low tercile	0.034	0.018	45.1%
Mid tercile	0.123	0.107	13.0%
High tercile	0.157	0.149	5.3%
Overall	0.104	0.091	12.5%

Table 3: GJP Year 1 Brier scores by dispersion tercile ($n = 94$ binary questions, mean panel size 895). The barycenter (= component-wise median) outperforms the arithmetic mean most strongly when the panel is cohesive.

The GJP binary finding adds an important dimension: the barycenter’s advantage holds with very large panels (~ 900 per question). Even with nearly a thousand forecasters, the median dominates the mean because it is robust to the minority who assign extreme probabilities.

5.2 Multi-Category Questions

The GJP also contains questions with $K \geq 3$ unordered outcomes — for example, “Who will be inaugurated as President of Russia in 2012?” (options: Medvedev, Putin, or Neither) or “Which candidate will win the Venezuelan presidential primary?” ($K = 5$). These questions require each forecaster to submit a full probability vector over K options; no natural ordering

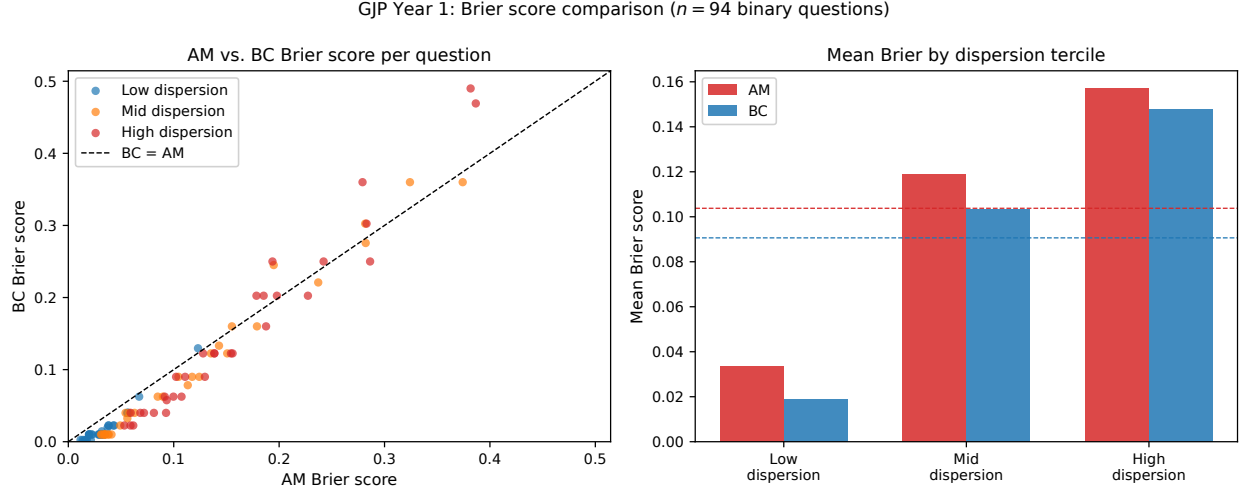


Figure 6: GJP Year 1: Brier score comparison for 94 binary questions. Left: scatter of AM vs. BC Brier score (one point per question; below the diagonal = BC wins; color indicates panel dispersion). Right: mean Brier score by dispersion tercile. The barycenter consistently outperforms the arithmetic mean.

exists among the outcomes. The indicator-cost ℓ^1 barycenter (Theorem 2.2) applies directly: the barycenter is the coordinatewise median of the expert probability vectors, providing a parameter-free aggregate that is robust to outlying opinions in any coordinate.

Data and protocol. We use all four years of the GJP [3], combining Years 1–4 (2011–2015). There are 116 closed multi-category questions with $K \in \{3, 4, 5\}$ across the four years ($K = 3$: 47 questions; $K = 4$: 50; $K = 5$: 19). For each question we take each forecaster’s last submitted forecast before closure, drop users with incomplete option coverage, and normalize rows to sum to one. We retain the 113 questions with at least 20 valid forecasters (median panel size: 379). We score each aggregate using the Brier score for K -category events, $BS = \sum_{k=1}^K (p_k - \mathbf{1}_{k=\text{outcome}})^2$.

Results. The ℓ^1 barycenter outperforms the arithmetic mean on 87 of 113 questions (77.0%; $p < 0.001$, Wilcoxon signed-rank test), with mean Brier scores of 0.303 (BC) versus 0.332 (AM), an 8.7% reduction (Table ?? and Figure ??). The advantage strengthens with K : the reduction is 3.1% for $K = 3$, 12.4% for $K = 4$, and 17.2% for $K = 5$, consistent

with Theorem 2.2’s prediction that with more outcomes the coordinatewise median has more dimensions along which to avoid outlier pull. The dispersion-conditional pattern from the binary setting and the SPF replicates here: in the low-dispersion half of questions (below-median mean squared distance to the AM), the barycenter wins 86% of questions with a 16.5% reduction; in the high-dispersion half, the win rate falls to 68% with a 4.3% reduction. This is precisely the regime structure predicted by Proposition 2.9: the indicator-cost barycenter’s coordinatewise robustness holds F^* near F_{true} in consensus panels, while the AM’s equal-weight averaging recovers ground when the panel is genuinely divided.

	n	AM Brier	BC Brier	BC improvement
<i>Panel A: By number of outcomes</i>				
$K = 3$	45	0.220	0.213	3.1%
$K = 4$	49	0.375	0.328	12.4%
$K = 5$	19	0.456	0.378	17.2%
<i>Panel B: By dispersion</i>				
Low dispersion	57	0.262	0.219	16.5%
High dispersion	56	0.403	0.388	4.3%
Overall	113	0.332	0.303	8.7%***

Table 4: GJP Years 1–4: ℓ^1 barycenter vs. arithmetic mean on 113 multi-category questions ($K \in \{3, 4, 5\}$). *** $p < 0.001$ (Wilcoxon). Dispersion split at median mean-squared distance of forecasters to the panel mean. The barycenter’s advantage grows with K and is concentrated in low-dispersion questions.

This result is the empirical grounding for Theorem 2.2: the ℓ^1 simplification is not merely a mathematical convenience but a practically useful aggregation rule for genuinely unordered categorical forecasts. Because the outcomes carry no ordering, the W_1 (ordered) barycenter and Hora et al.’s CDF-median are not applicable; the indicator-cost ℓ^1 barycenter is the natural and novel choice for this setting.

6 Application to the Survey of Professional Forecasters

The opioid survey (Appendix B) treats the 20 interventions as an *unordered* outcome space, so the indicator distance is the natural choice. Many expert-aggregation problems, however,

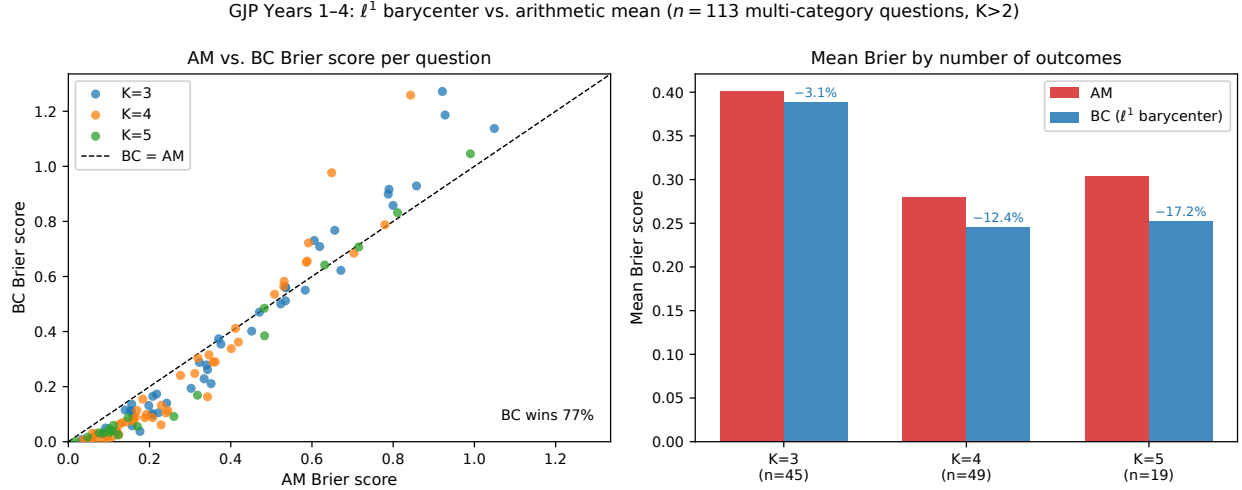


Figure 7: GJP Years 1–4: ℓ^1 barycenter vs. arithmetic mean on 113 multi-category questions. Left: scatter of AM vs. BC Brier score (below the diagonal = BC wins; color indicates K). Right: mean Brier score by K . The improvement over the arithmetic mean grows with the number of outcomes.

involve *ordered* outcomes: survey respondents assign probabilities to ordered bins. For such data the W_1 barycenter has the closed-form characterization derived in Proposition 2.5: it equals the component-wise median of the empirical CDFs. This section illustrates both the practical convenience of that formula and its robustness implications using data from the Survey of Professional Forecasters (SPF).

6.1 Data

The Philadelphia Federal Reserve Bank has administered the SPF every quarter since 1968. In each survey, professional forecasters submit probabilistic predictions for macroeconomic variables by allocating probability mass across a fixed set of ordered bins. We use the *core CPI* probability question (variable `PRCCPI`), which asks each forecaster to assign probabilities to 10 ordered bins for fourth-quarter-over-fourth-quarter core CPI inflation in the current and following year. The 10 bins are (in ascending order): decline, 0.0–0.4%, 0.5–0.9%, 1.0–1.4%, 1.5–1.9%, 2.0–2.4%, 2.5–2.9%, 3.0–3.4%, 3.5–3.9%, and $\geq 4.0\%$.

We focus on the 2022:Q1 survey’s *next-year* histograms, each forecaster’s probability

distribution for 2023 core CPI inflation, using the $n = 27$ respondents who provided a complete histogram summing to 100%. This survey was conducted in February 2022, when realized core CPI was running near 6% and forecasters disagreed sharply about how quickly inflation would moderate.

6.2 Aggregate Distributions

Figure 7 displays the 27 individual histograms as a heatmap. The heterogeneity is striking: some forecasters concentrated nearly all mass on 2–3% (expecting a rapid return to target), while others spread mass across higher bins. A few placed non-trivial probability on the decline and $\geq 4\%$ tail bins.

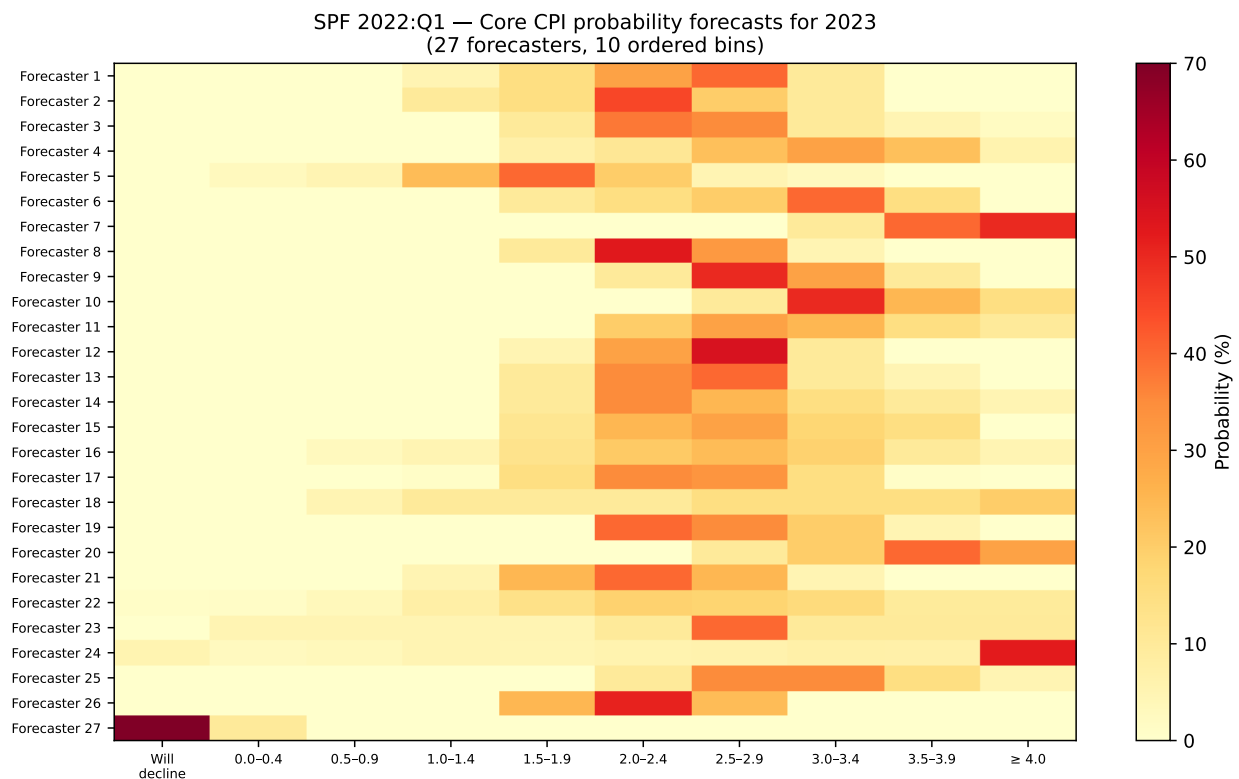


Figure 8: SPF 2022:Q1: probability distributions for 2023 core CPI inflation across 27 professional forecasters. Each row is one forecaster’s histogram over 10 ordered bins (ascending from “will decline” to $\geq 4\%$). Cell color encodes probability mass (darker = higher).

Because the outcomes are ordered, we use the W_1 distance (absolute rank difference) as

the ground cost. This respects the ordinal structure of the inflation bins: shifting probability from the 2–3% bin to the 3–4% bin incurs less transport cost than shifting it to the deflation bin, correctly reflecting the asymmetry in the outcomes. By Proposition 2.5, the W_1 barycenter equals the PMF obtained by component-wise CDF median. We verified this numerically: the closed-form formula and the LP solver agree to machine precision (ℓ^1 discrepancy $< 10^{-4}$).

Figure 8 compares the arithmetic mean and the W_1 barycenter. The differences are substantial. The arithmetic mean places 3.6% mass on the “decline” bin and 8.1% on the $\geq 4\%$ bin, for a combined tail probability of 11.7%. The barycenter assigns *zero* mass to both tails. Instead, it concentrates 60% of its mass on the 2.0–2.9% range and 30% on 3.0–3.9%, reflecting the view shared by a majority of forecasters. The arithmetic mean disperses mass across all bins because a minority of extreme views pull it toward the tails.

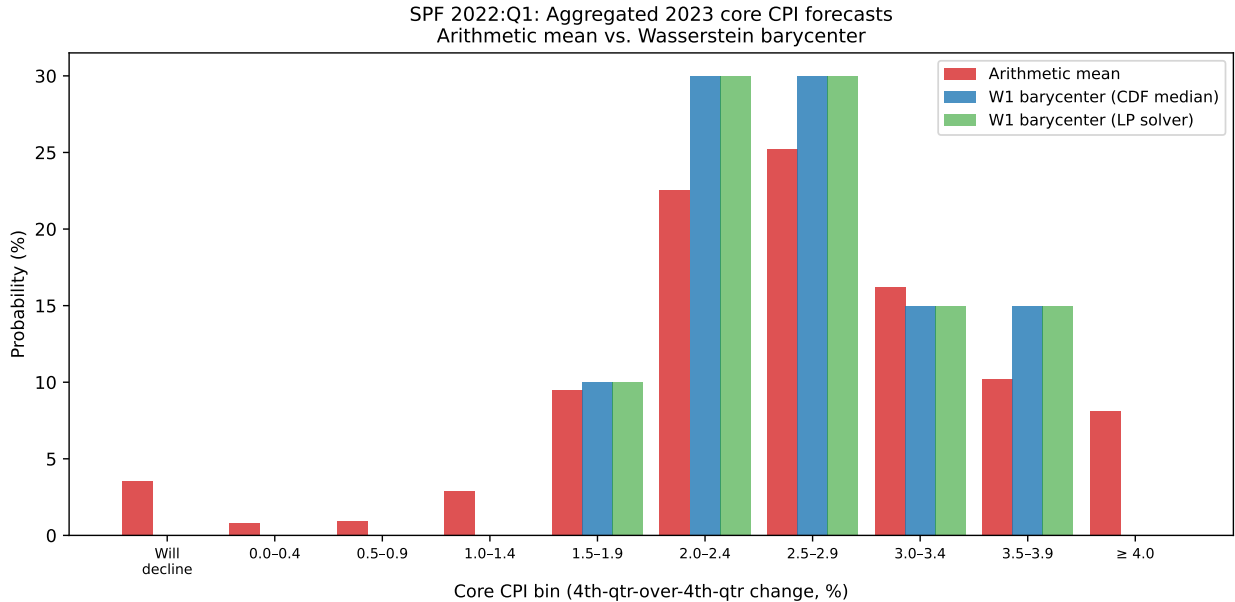


Figure 9: Arithmetic mean (red), W_1 barycenter via CDF-median formula (blue), and W_1 barycenter via LP solver (green) for 2023 core CPI, SPF 2022:Q1. The closed-form and LP results coincide exactly.

Figure 9 shows the individual CDFs (grey), the arithmetic-mean CDF (red), and the barycenter CDF (blue). The barycenter CDF is a step function whose steps track the *vertical*

589 *median* of the individual CDFs at each bin boundary. The AM CDF is smoother but pulled
590 outward toward the tails by outlying forecasters.

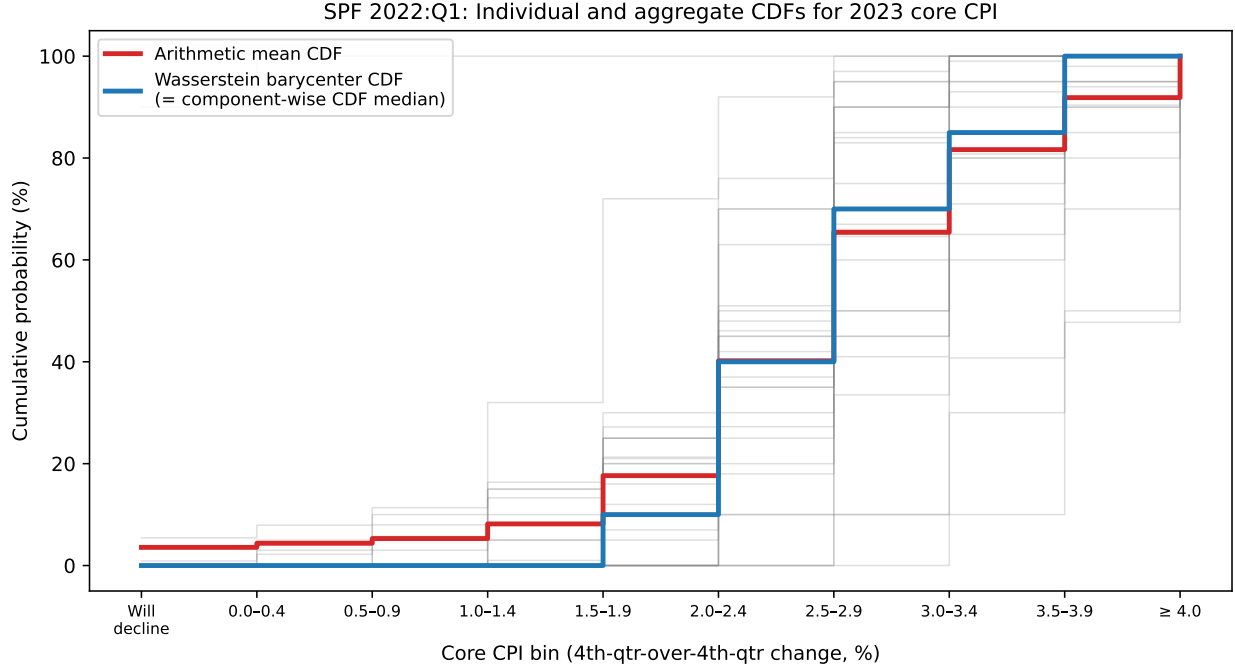


Figure 10: Individual CDFs (grey, $n = 27$) and aggregate CDFs for the arithmetic mean (red) and the W_1 barycenter (blue). The barycenter CDF is the component-wise median of the individual CDFs, confirming Proposition 2.5.

591 6.3 Historical Forecast Evaluation

592 Evaluation protocol

593 **Forecast variable.** We use the current-year PRCCPI histograms (columns PRCCPI1–
594 PRCCPI10), in which each respondent distributes 100 probability points across 10 ordered
595 bins for Q4/Q4 core CPI inflation in the current calendar year. The horizon is current-year
596 (not next-year); we choose this horizon because the realized value is unambiguously assigned
597 within the same calendar year and the panel is largest.

598 **Sample construction.** The sample covers all survey quarters from 2007:Q1 through
599 2025:Q4. A respondent’s row is retained if its bin probabilities sum to within 2 percentage
600 points of 100; retained histograms are renormalized to exactly 1 and reversed to ascending

bin order. Quarters with fewer than five valid respondents are excluded, leaving $n = 76$ usable quarters.

Realized outcomes. Each realized value is the December-over-December percent change in seasonally adjusted core CPI (FRED series `CPILFESL`) for the forecast year. The SA and NSA series are nearly identical for this variable. Each realized value is mapped to one of the 10 bins using the boundaries in Table 4.

Scoring rules. The primary scoring rule is the Ranked Probability Score (RPS; see Remark 2.9) [16]: lower RPS is better. The secondary criterion is the probability mass the aggregate places on the realized bin (mass score); higher is better. While not a proper scoring rule, the mass score provides an interpretable measure of concentration at the realized outcome and complements RPS, which penalizes dispersion across the full CDF. The two criteria together distinguish methods that concentrate near the correct outcome from those that spread probability broadly.

Inference. Score differences are tested via two complementary procedures. One-sided Wilcoxon signed-rank tests [16] assess whether a method’s RPS is stochastically smaller than the AM’s (robust to non-normality and heavy tails). Diebold-Mariano (DM) tests [25] assess whether the mean loss differential is zero, using Newey-West HAC standard errors to account for serial dependence. The two tests are complementary: Wilcoxon measures *consistency* of the advantage (win rate), while DM measures its *average magnitude*. When the loss differential is skewed, as in 2021–2022 where both methods suffered large losses, the tests can diverge, and we report both.

Calibration. We assess calibration via bin-level reliability diagrams: for each of the 10 bins and each method, we plot the mean predicted probability against the empirical frequency of that bin being realized. Well-calibrated forecasts lie on the 45-degree line. Results are reported in Figure 14.

The single-quarter illustration above motivates a question of practical importance: does the W_1 barycenter produce better probabilistic forecasts than the arithmetic mean when

628 evaluated against realized outcomes?

629 **Overall results.** We compare four methods: the arithmetic mean (AM), the logarithmic
630 opinion pool (LP), the symmetrically trimmed arithmetic mean (TM, 10% trim), and the W_1
631 barycenter (BC). Panel A of Table 4 reports both metrics over the full 76-quarter sample.
632 The TM achieves the lowest mean RPS among the four methods (0.435, wins 76.3% vs. AM,
633 Wilcoxon $p < 0.001$, DM $p = 0.002$), with the BC close behind (0.439, wins 68.4%, Wilcoxon
634 $p = 0.005$). The Wilcoxon and DM tests tell a complementary story for BC vs. AM: Wilcoxon
635 confirms BC wins more often, while DM $p = 0.225$ indicates the *average magnitude* of the
636 gain is not individually significant, reflecting the fact that 2021–2022 inflation-surge quarters
637 inflated the variance of the loss differential and attenuated the mean gain. The dispersion-
638 conditional DM tests resolve this: in low-dispersion quarters both tests agree strongly (DM
639 $p < 0.001$, Wilcoxon $p < 0.001$), confirming the BC’s structural advantage precisely where
640 the theory predicts it. The apparent RPS gap between TM and BC (0.435 vs. 0.439) is
641 not statistically significant by either test (Wilcoxon $p = 0.82$, DM $p = 0.63$): TM does
642 not reliably outperform the BC; its narrow average advantage disappears in the adaptive
643 blend. The LP has the worst RPS (0.486, DM $p = 0.95$) despite the highest P(correct bin)
644 (0.491 vs. 0.424 for BC). This divergence (LP is best at placing mass on the realized bin yet
645 worst at RPS) reflects the ℓ^1 – ℓ^2 mismatch of Remark 2.9: the LP’s extreme concentration
646 minimizes ℓ^1 error at the mode, but its near-zero tail mass amplifies squared CDF error in the
647 increasingly many quarters where a tail bin is realized. The BC balances both objectives: it
648 places substantially more mass on the realized bin than the AM (75.0% win rate, $p < 0.001$)
649 while preserving more tail coverage than the LP.

650 **Dispersion-conditional results.** The theory makes a sharp prediction about when the
651 barycenter advantage is largest. Corollary 2.8 shows the W_1 barycenter is locally constant
652 when panel CDFs are separated from their neighbors by a positive margin: low cross-sectional
653 dispersion corresponds to tightly clustered CDFs with large median margins, so the barycen-

ter is most stable and most concentrated on the consensus precisely when the panel broadly agrees. We therefore test a pre-specified, theory-driven partition: quarters in which the panel agrees (low dispersion) versus quarters in which it is genuinely divided (high dispersion). Panel B splits the sample at the median cross-sectional variance into 38 low- and 38 high-dispersion quarters. The probability-on-realized-bin metric shows the barycenter outperforming in *both* groups: 0.496 versus 0.430 in low-dispersion quarters, and 0.353 versus 0.303 in high-dispersion quarters. The RPS results are more nuanced: the barycenter wins decisively in low-dispersion quarters (mean RPS 0.301 vs. 0.328, $p < 0.001$), while the arithmetic mean is marginally better in high-dispersion quarters (mean RPS 0.567 vs. 0.578, $p = 0.37$). The divergence between the two metrics in high-dispersion quarters reflects the ℓ^1 – ℓ^2 mismatch identified in Remark 2.9: the barycenter minimizes an ℓ^1 CDF objective, so it concentrates near the panel median; but RPS penalizes squared CDF deviations, and the AM’s broader CDF is geometrically closer to the step CDFs of tail outcomes. The barycenter still places more mass on the realized bin even in high-dispersion quarters, but the AM’s retained tail mass yields smaller cumulative squared deviations when an extreme outcome materializes. The 2021–2022 inflation surge is the clearest illustration: both methods underestimated realized core CPI ($\geq 4\%$), but the AM retained more tail mass, reducing its RPS deficit in those extreme quarters.

In low-dispersion quarters the economic interpretation is straightforward: the panel agrees, the barycenter correctly concentrates mass on the consensus region, and the consensus is usually right. In high-dispersion quarters the panel is genuinely uncertain; the arithmetic mean’s equal weighting preserves more tail mass, which pays off under RPS when an extreme outcome materializes, even though the barycenter still places more mass on the correct bin in expectation.

Figures 10–13 provide the visual summary. Figure 10 shows the probability mass assigned to the correct bin in each quarter; the barycenter’s advantage is visible throughout the sample and in both dispersion groups. Figure 11 shows the RPS time series for both methods.

Table 5: SPF current-year backtest: four-method comparison, 2007:Q1–2025:Q4. AM = arithmetic mean; LP = logarithmic opinion pool; TM = symmetrically trimmed mean (10%); BC = W_1 barycenter. “P(correct bin)”: mean probability on the realized bin (higher is better). RPS: mean Ranked Probability Score (lower is better). Wilcoxon: one-sided signed-rank test vs. AM (H_1 : lower RPS). DM: Diebold-Mariano test [25] with Newey-West HAC standard errors vs. AM. Adaptive blend results are from the 56-quarter OOS evaluation (Section 6.4).

Method	P(correct bin)	Mean RPS	Wins% vs AM	Wilcoxon p	DM p
<i>Panel A: Full sample ($n = 76$ quarters)</i>					
Arithmetic mean (AM)	0.367	0.447	—	—	—
Log pool (LP)	0.491	0.486	57.9%	0.886	0.952
Trimmed mean (TM, 10%)	0.393	0.435	76.3%	0.0001	0.002
W_1 Barycenter (BC)	0.424	0.439	68.4%	0.005	0.225
Adaptive blend (OOS)	—	0.404	—	<0.001	—
<i>Panel B: By cross-sectional dispersion ($n = 38$ each)</i>					
	Low dispersion		High dispersion		
	P(corr. bin)	RPS	P(corr. bin)	RPS	
AM	0.430	0.328	0.303	0.567	
BC	0.496	0.301	0.353	0.578	
<i>DM p-value (BC vs AM, low/high disp): <0.001 / 0.71</i>					

Figure 12 plots the quarter-by-quarter RPS difference, with the high-dispersion quarters (shaded) containing the episodes where the arithmetic mean recovers ground. Figure 13 summarizes the dispersion-conditional RPS comparison.

6.4 Adaptive Blending

The historical results suggest a natural extension. The W_1 barycenter improves on the arithmetic mean on average, but the arithmetic mean retains value in high-dispersion quarters under the RPS. A convex blend

$$\hat{\Psi}_t(\lambda) = \lambda \Psi_t^* + (1 - \lambda) \bar{P}_t, \quad \lambda \in [0, 1],$$

interpolates between the two extremes. At $\lambda = 1$ this is the barycenter; at $\lambda = 0$ it is the arithmetic mean.

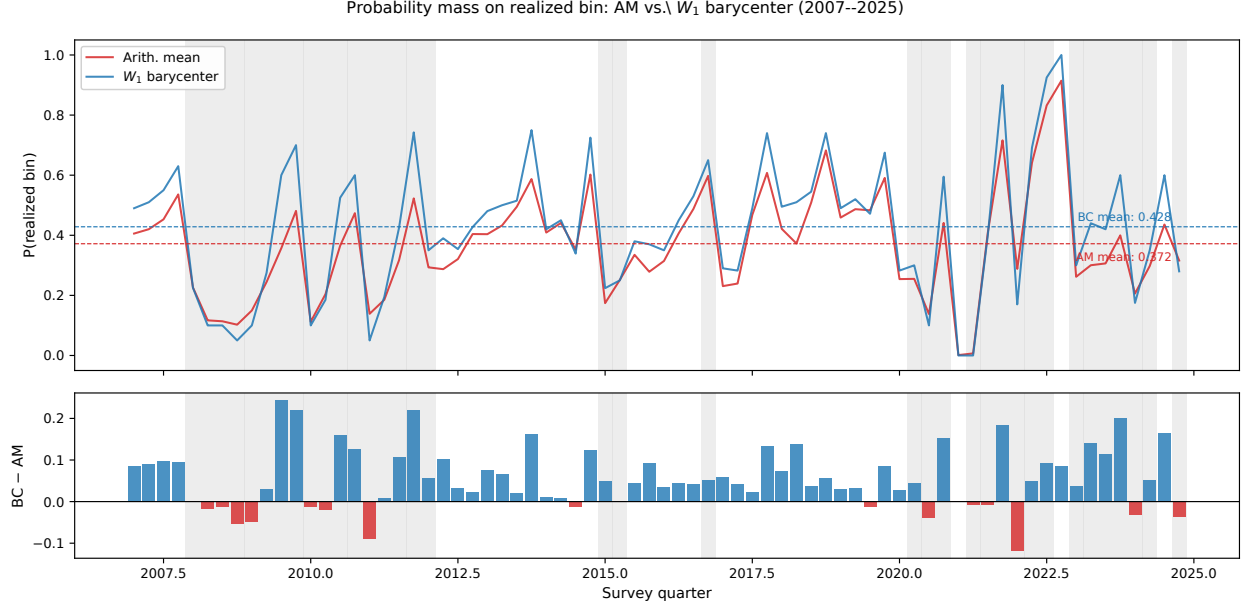


Figure 11: Top: probability mass assigned to the realized outcome bin by the arithmetic mean (red) and W_1 barycenter (blue), 2007:Q1–2025:Q4. Dashed lines mark the sample means (AM: 0.367; BC: 0.424). Shaded regions are high-dispersion quarters. Bottom: quarter-by-quarter difference (BC – AM); positive bars (blue) indicate quarters where the barycenter places more mass on the correct bin.

Oracle. A grid search over $\lambda \in \{0, 0.005, \dots, 1\}$ using all 76 quarters identifies $\lambda^* = 0.70$ as the mean-RPS minimizer, yielding an in-sample mean RPS of 0.437, lower than both the arithmetic mean (0.447) and the pure barycenter (0.439). The improvement is modest but reveals that neither extreme is optimal.

Out-of-sample evaluation. To avoid look-ahead bias we apply an expanding-window protocol: for each quarter t (starting from quarter 21) we estimate $\hat{\lambda}_t$ by minimizing mean RPS over all prior quarters and apply it only to quarter t . In a richer adaptive version, $\hat{\lambda}_t = \sigma(a_t + b_t D_t)$ where D_t is the cross-sectional variance of individual forecast means and (a_t, b_t) are estimated by Nelder-Mead on the expanding window. Results over the 56-quarter out-of-sample period are reported in Table 5 and Figure 15.

The adaptive blend achieves a mean RPS of 0.404, a 4.3% improvement over the arithmetic mean ($p < 0.001$) and a 1.6% improvement over the unweighted barycenter. The

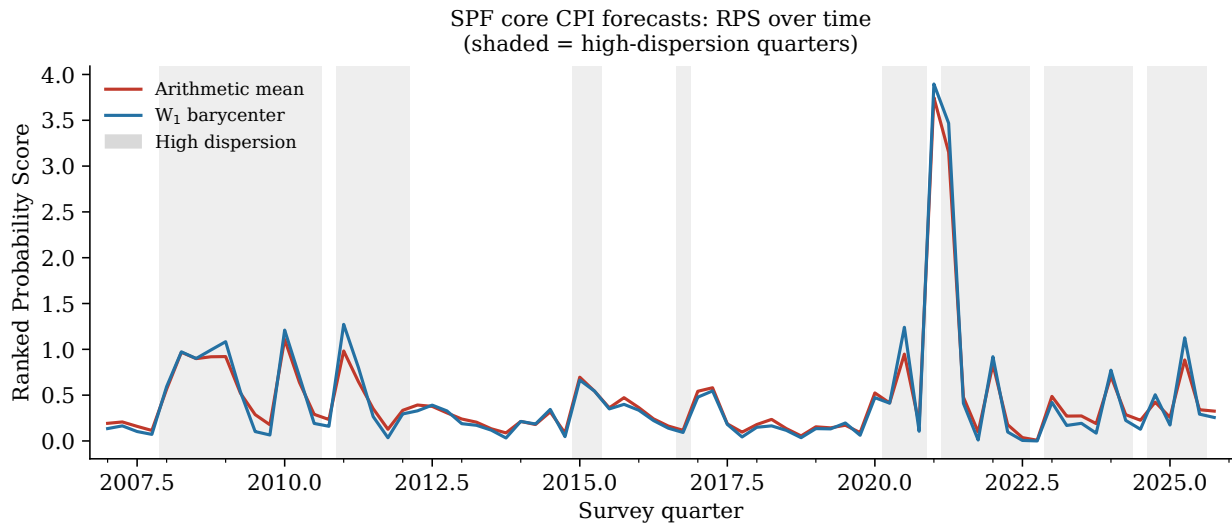


Figure 12: AM (red) and W_1 barycenter (blue) RPS over 76 survey quarters, 2007:Q1–2025:Q4. Shaded regions mark high-dispersion quarters (cross-sectional variance of forecast means above the sample median).

Method	Mean RPS	p -value (vs. AM)	Wins vs. AM
Arithmetic mean	0.423	—	—
W_1 barycenter	0.411	0.002	75%
Trimmed mean	0.409	<0.001	80%
Fixed blend ($\hat{\lambda} \approx 0.71$)	0.413	0.001	79%
Adaptive blend	0.404	<0.001	64%

Table 6: Out-of-sample RPS comparison (56 quarters, expanding-window evaluation). p -values from Wilcoxon signed-rank test versus the arithmetic mean. The adaptive blend achieves the lowest mean RPS; its margin over the trimmed mean (0.409 vs. 0.404) is not statistically significant ($p = 0.30$, Wilcoxon).

702 trimmed mean achieves an OOS RPS of 0.409; the blend’s 1.1% margin over TM is not
703 statistically significant ($p = 0.30$, Wilcoxon), so the two methods are approximately tied on
704 this evaluation window. The expanding-window estimate of λ converges rapidly to approx-
705 imately 0.71, closely matching the oracle value, which indicates that the optimal mixture
706 weight is stable over time and requires no large calibration sample.

707 6.5 Replication on the ECB Survey of Professional Forecasters

708 Taylor [14] applies depth-based median aggregation to ECB SPF inflation forecasts using
709 a partially overlapping sample, providing an independent evaluation of robust CDF aggre-

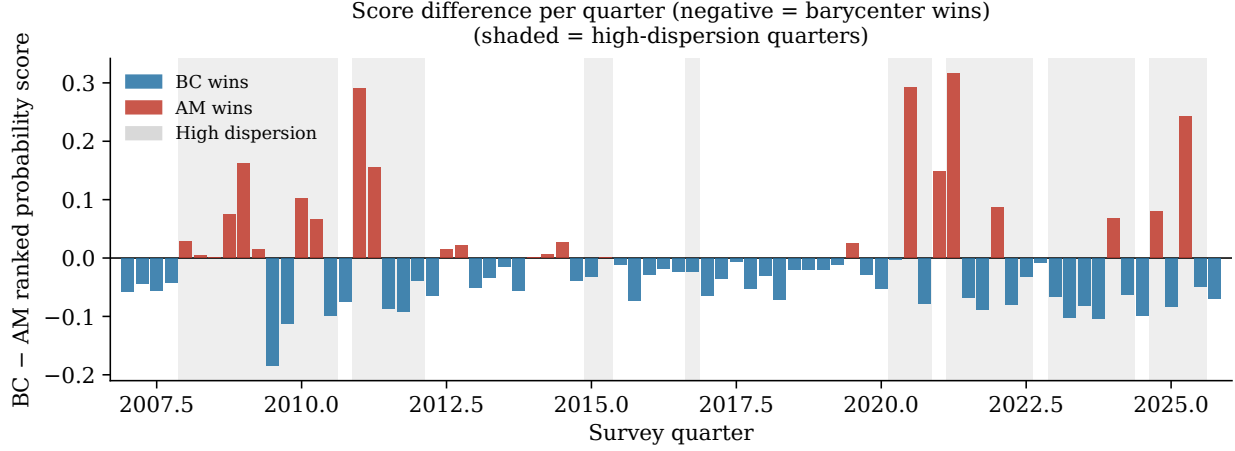


Figure 13: Quarter-by-quarter RPS difference (BC – AM). Negative bars (blue) indicate quarters where the barycenter achieves a lower (better) RPS; positive bars (red) indicate quarters where the arithmetic mean wins. The barycenter outperforms in 68.4% of quarters.

gation in the same institutional setting. Our estimator (the W_1 barycenter, equivalently the vertical CDF median) differs from Taylor’s halfspace-depth median in construction but shares the same qualitative motivation. The contribution of our ECB evaluation relative to Taylor [14] is threefold: (i) we use a simpler, parameter-free estimator; (ii) our sample extends through 2024Q3, covering the 2022–2023 Euro-area inflation surge; and (iii) we document the dispersion-conditional pattern — the mechanism linking the ℓ^1 – ℓ^2 mismatch to empirical regimes — which is not reported in Taylor.

To assess whether the US SPF findings are specific to that survey or reflect a more general property of barycenter aggregation, we replicate the backtest on the ECB Survey of Professional Forecasters (ECB SPF), which has been conducted quarterly since 1999 [1]. The ECB SPF covers the Euro area and asks respondents to assign probabilities to ordered bins for headline HICP inflation, real GDP growth, and the unemployment rate. We use the headline HICP current-year forecasts over the period 2010Q1–2024Q3 ($n = 56$ quarter-surveys), during which the bin structure is stable at 12 ordered bins of width 0.5 pp from below -1% to above 4% . Realized values are annual-average HICP YoY from the ECB Statistical Data Warehouse. The evaluation protocol (AM, W_1 barycenter, RPS scoring, dispersion split) is identical to Section 6.3.

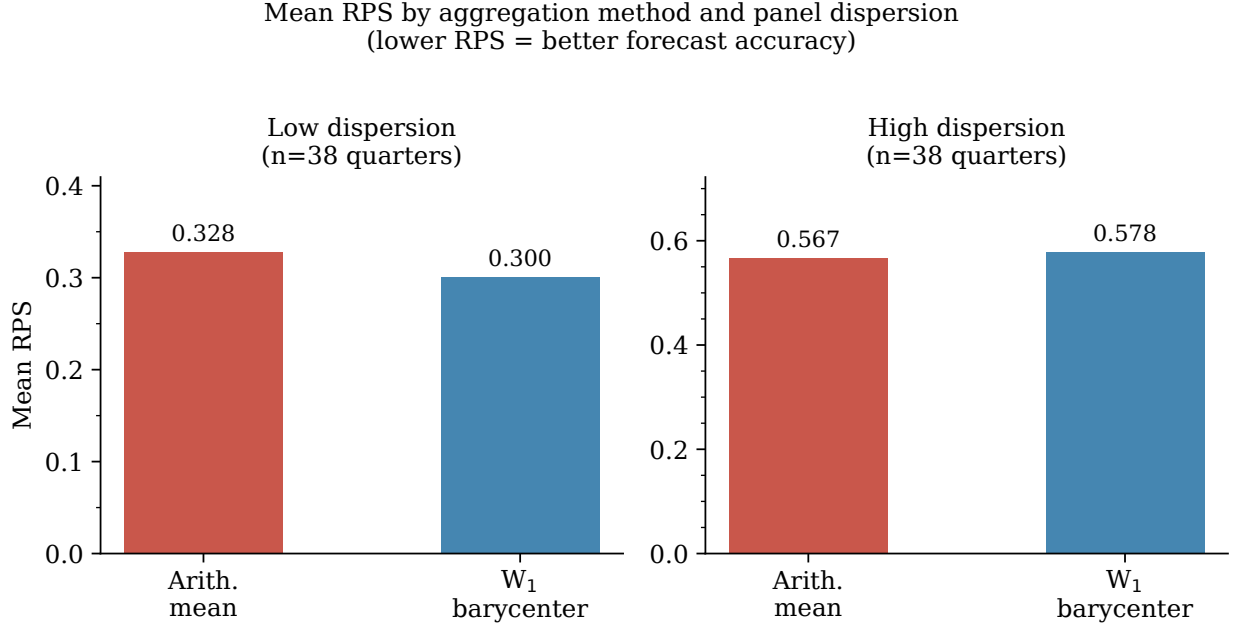


Figure 14: Mean RPS by aggregation method and dispersion group. In low-dispersion quarters the barycenter reduces mean RPS by 8.3% relative to the arithmetic mean; in high-dispersion quarters the arithmetic mean is marginally better.

Results. The barycenter assigns more probability mass to the realized outcome bin in 78.6% of quarter-surveys ($p < 0.001$, Wilcoxon), with a mean mass of 0.457 versus 0.430 for the arithmetic mean, a 6.3% improvement (Figure 16). This replicates the core finding from the US SPF on an independent panel in a different institutional setting and with a different realized variable.

The aggregate RPS advantage is near zero overall (mean RPS 0.441 AM vs. 0.441 BC, Wilcoxon $p = 0.23$), but this masks a clean dispersion-conditional pattern that precisely replicates the US SPF finding. In low-dispersion quarters ($n = 28$), the barycenter achieves meaningfully lower RPS (mean 0.255 vs. 0.266), and this advantage is statistically significant by both tests (Wilcoxon $p = 0.012$, DM $p = 0.006$). In high-dispersion quarters the arithmetic mean wins marginally (mean 0.616 vs. 0.626), with no significant difference (Wilcoxon $p = 0.57$, DM $p = 0.89$). The overall near-tie is explained by the 2021–2023 Euro-area inflation surge: annual-average HICP reached 8.4% in 2022, well above the highest bin boundary ($\geq 4\%$), an outcome that both methods failed to anticipate but that the arithmetic mean

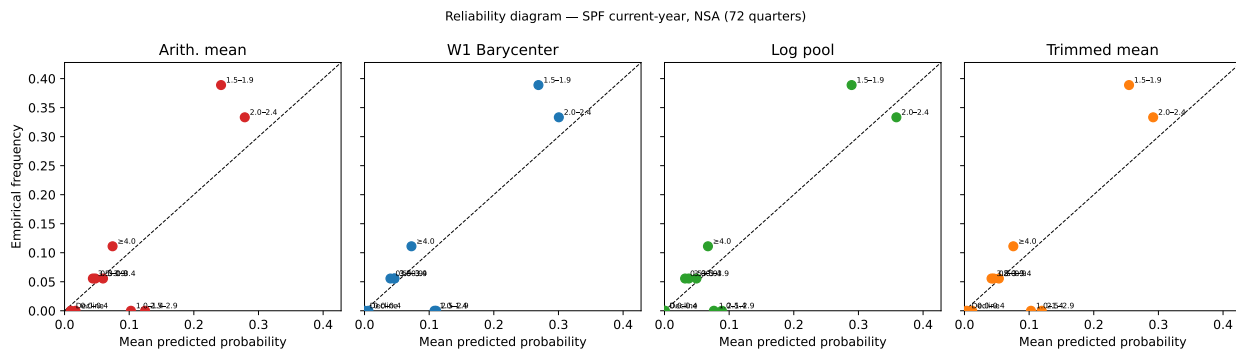


Figure 15: Reliability diagrams for all four aggregation methods over 76 SPF current-year quarters. Each point represents one of the 10 bins; the horizontal axis is the method’s mean predicted probability for that bin and the vertical axis is the empirical frequency with which that bin was realized. Points on the dashed 45-degree line indicate perfect calibration. All four methods are approximately calibrated, with no systematic over- or under-confidence.

handles less badly under RPS by retaining more tail mass. These extreme quarters dominate the high-dispersion subsample and attenuate the aggregate average. The replication finding is therefore not a null result but a confirmation: the dispersion-conditional pattern holds on an independent panel with a different variable, a different bin structure, and a different institutional setting.

Method	Mean score		
	RPS	P(correct bin)	Wins (mass)
Arithmetic mean	0.441	0.430	—
W_1 barycenter	0.441	0.457	78.6%***
<i>Panel B: By dispersion ($n = 28$ each)</i>			
	Mean RPS		
	Low disp.	High disp.	
Arithmetic mean	0.266	0.616	
W_1 barycenter	0.255	0.626	

Table 7: ECB SPF backtest (56 quarter-surveys, 2010Q1–2024Q3, current-year headline HICP). *** Wilcoxon $p < 0.001$. Overall RPS difference is not significant ($p = 0.23$).

Interpretation. The ECB SPF replication confirms that the barycenter’s concentration near the panel consensus is a reliable property, not a sample artifact: it places more mass on the realized bin in both the US and Euro-area panels. The non-significant RPS result for the ECB sample reflects the particular severity of the 2022 inflation episode, whose outcome

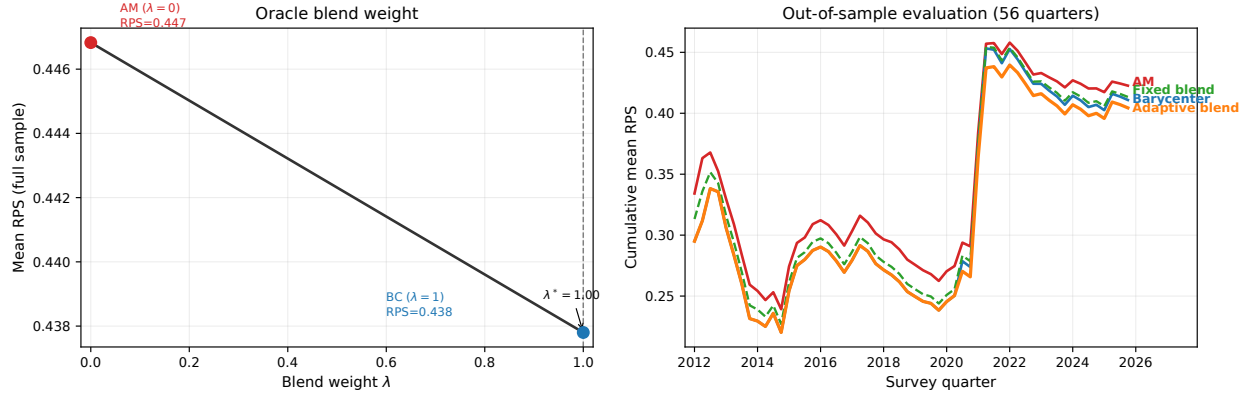


Figure 16: Left: oracle mean RPS as a function of blend weight λ (full sample). The minimum at $\lambda^* = 0.70$ lies between the arithmetic mean ($\lambda = 0$) and the barycenter ($\lambda = 1$). Right: cumulative out-of-sample mean RPS for the arithmetic mean, W_1 barycenter, fixed blend, and dispersion-adaptive blend over the 56-quarter evaluation period.

lay far outside the normal support of the distribution. Taken together, the two backtests suggest that the barycenter gains from sharper concentration near the consensus and can lose from reduced tail mass in extreme-realization quarters.

7 Discussion

This paper proposes the Wasserstein barycenter as a robust method for aggregating expert probability forecasts over finite outcome spaces. The central theoretical result is Theorem 2.2: under the indicator ground cost, the barycenter reduces to a constrained ℓ^1 aggregation problem. This simplification has two important consequences. First, it makes computation straightforward: the problem is a convex linear program in NK variables, solvable by standard LP solvers; for ordered spaces, Proposition 2.5 gives a fully closed-form answer (component-wise CDF median) with no solver required. Second, it identifies the barycenter as a median-type aggregate: the unconstrained minimizer of the ℓ^1 objective assigns the coordinatewise median to each state, and the simplex constraint shifts this toward a normalized version of the median. For ordered outcome spaces the closed form is exact: the W_1 barycenter is the distribution whose CDF is the component-wise median of the indi-

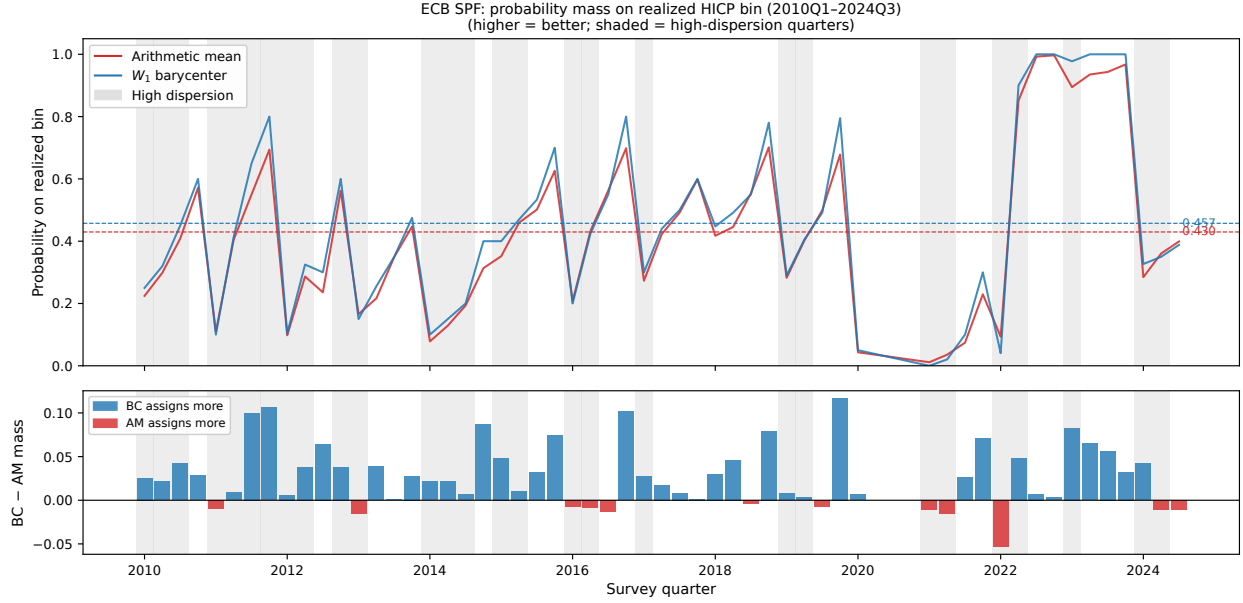


Figure 17: ECB SPF: probability mass assigned to the realized annual-average HICP bin by the arithmetic mean and W_1 barycenter, 2010Q1–2024Q3. The barycenter consistently assigns more mass to the correct bin. Shaded quarters are high-dispersion (above-median cross-sectional variance). The 2022 inflation surge (realized HICP 8.4%) is visible as a period of low mass for both methods.

vidual CDFs (Proposition 2.5), a result verified numerically to machine precision in the SPF application.

When the barycenter wins and when it does not. The barycenter is not a uniformly superior alternative to the arithmetic mean, and we do not claim it is. In high-dispersion quarters, when the panel is genuinely divided and tail outcomes are plausible, the arithmetic mean’s broader CDF yields lower RPS, because it retains the tail coverage that squared-CDF scoring rewards when an extreme bin is realized. The trimmed mean achieves similar overall RPS to the barycenter (the gap is not statistically significant by either Wilcoxon or DM tests), and it remains a reasonable choice when a simple, interpretable aggregate is needed. The barycenter’s advantage is concentrated in low-dispersion quarters, where the theoretical regime structure (Corollary 2.8) predicts it: tightly clustered CDFs imply large median margins, so the aggregate is locally constant and concentrates mass precisely where the panel agrees. The adaptive blend is preferred over any fixed method when a calibration

778 window is available, because it does not commit to either regime.

779 **Robustness interpretation.** The EIF analysis formalizes the intuition that the barycen-
780 ter is less sensitive than the arithmetic mean to extreme opinions. The AM gives every
781 expert equal weight regardless of distance from the consensus; the barycenter penalizes out-
782 lying opinions through transport cost, so an expert far from the current aggregate exerts
783 smaller influence. Empirically, in 90% of simulation replicates the barycenter assigns less
784 probability to an outlier’s preferred state than the AM. The tradeoff is efficiency: under a
785 well-specified homogeneous expert population, the AM has lower MSE by a factor of approx-
786 imately 1.4–1.9.

787 **Forecast evaluation.** The historical backtest in Section 6.3 provides direct evidence that
788 the theoretical properties of the barycenter translate into empirical gains. The barycenter
789 assigns more probability mass to the realized outcome bin in a majority of quarters, with a
790 mean improvement of 15.6% (0.424 vs. 0.367), and achieves a lower average Ranked Prob-
791 ability Score than the arithmetic mean. The advantage is concentrated in low-dispersion
792 quarters; the arithmetic mean recovers ground under RPS in high-dispersion quarters be-
793 cause its broader tail coverage reduces cumulative CDF deviations when extreme outcomes
794 materialize. A dispersion-adaptive convex blend of the two methods achieves the lowest out-
795 of-sample RPS (0.404), versus 0.409 for the trimmed mean, 0.411 for the pure barycenter,
796 and 0.423 for the arithmetic mean. The blend’s margin over the trimmed mean is small
797 (1.1%) and not statistically significant ($p = 0.30$); both the blend and TM significantly out-
798 perform the arithmetic mean ($p < 0.001$ and $p < 0.001$ respectively). The optimal mixture
799 weight ($\lambda^* \approx 0.70$) is stable over time, matching the oracle value within the expanding-
800 window evaluation. These results together characterize when each method is preferred: the
801 barycenter is the better single aggregate when the panel is broadly in agreement, while the
802 arithmetic mean’s equal weighting preserves tail mass that is occasionally valuable when
803 extreme outcomes materialize, and the blend dominates both by adapting continuously. The

Wasserstein barycenter is not calibrated by construction: calibration is a property of the individual expert forecasts and is not in general preserved under nonlinear combination. The reliability diagrams (Figure 14) confirm no systematic miscalibration across methods; all four aggregates track the 45-degree line without evidence of over- or under-confidence. The comparison against the logarithmic pool and the trimmed mean (Table 4) reveals that the ℓ^1 – ℓ^2 mismatch between the pooling objective and RPS is not specific to the barycenter: the LP, which concentrates mass even more aggressively, achieves the highest P(correct bin) but the worst RPS. The trimmed mean, which excises extreme forecasters symmetrically, achieves the lowest single-method RPS. Extending the evaluation to multi-horizon forecasts, other outcome variables, and extremization [20] are natural further directions.

Remark 12 (Practical guidance). *The primary recommendation is the dispersion-adaptive convex blend (Section 6.4): it requires no decision about which regime applies, achieves the lowest out-of-sample RPS of any method evaluated, and automatically adjusts to the panel’s cross-sectional agreement. When a single non-adaptive aggregate is needed, the choice is guided by dispersion. When cross-sectional dispersion is low (the panel is broadly in agreement), the W_1 barycenter is preferred: it concentrates mass on the consensus region and achieves lower RPS than both AM and LP, with the advantage significant by both Wilcoxon and Diebold-Mariano tests. When dispersion is high (the panel is genuinely divided and tail outcomes are plausible), the arithmetic mean or trimmed mean preserves the tail coverage needed for strong RPS performance under extreme realizations. The blend operationalizes this transition continuously rather than as a binary switch.*

Applications and extensions. The applications illustrate the method on both ordered (SPF, W_1 distance) and unordered (opioid survey, indicator distance; Appendix B) outcome spaces. A natural extension is to weighted barycenters, in which experts receive unequal weights reflecting their track record or domain expertise. A natural population model for expert probability vectors is the Dirichlet distribution; however, aggregation via Dirichlet

MLE does not generally satisfy boundedness or unanimity, and therefore differs conceptually from the deterministic aggregation rules studied here. The extension to probability distributions over network-valued outcomes (Appendix A) illustrates that the framework applies to any finite metric space without modification. The multivariate case, in which experts provide joint distributions over multiple outcomes, raises new questions about the structure of the barycenter that we leave for future work.

A Barycenters for Network-Valued Outcomes

The framework extends naturally to settings where the outcome space is a set of network structures [26]. Let $\mathcal{Y}_n$ denote the space of binary adjacency matrices on n nodes: the set of $n \times n$ zero-diagonal matrices with entries in $\{0, 1\}$, equipped with the Hamming distance

$$d(Y, Y') = \sum_{m \neq n} |Y_{mn} - Y'_{mn}|.$$

A probability mass function over $\mathcal{Y}_n$ assigns probability $p_j(Y)$ to each network $Y \in \mathcal{Y}_n$. Existence of the barycenter follows from the compactness argument of Theorem 2.2 applied to the finite metric space $(\mathcal{Y}_n, d)$. The full-diagonal criterion (Proposition C.1) carries over: in any optimal solution, $\Pi_Y^{(j)} = \min\{p_j(Y), \Psi(Y)\}$ for each Y .

Unlike the unordered case where all off-diagonal pairs are equidistant, pairs of networks differing in k edges incur cost k under W_1 (or k^2 under W_2). After diagonal entries are maximized, remaining mass is assigned to off-diagonal entries in order of increasing Hamming distance.

The result extends to integer-valued (weighted) networks by replacing $\{0, 1\}$ with $\{0, \dots, M\}$ and using the ℓ^1 Hamming distance.

B Application to a Survey of Public Health Experts

We apply the methods developed above to a survey conducted by the *New York Times* in which 30 public health experts were each asked to allocate a hypothetical budget across 20 policy interventions aimed at addressing the opioid crisis. Each expert’s allocation is a probability vector over the 20 interventions and the collection of 30 vectors constitutes a natural instance of the expert aggregation problem.

B.1 Data

The 20 interventions span four policy categories: *treatment* (medication-assisted treatment, medicaid expansion, addiction treatment in prisons, drug courts, research); *harm reduction* (naloxone, needle exchanges, supervised consumption spaces, HIV/HCV treatment, fentanyl testing strips, medical examiners); *prevention* (community development, education, pain research, post-incarceration social programs); and *supply* (interdiction, prescription drug monitoring, local police forces, border wall, reducing diversion). Expert allocations range from near-zero (many experts allocate nothing to border-wall or supervised consumption spaces) to over 50% (one expert allocates half their budget to medication-assisted treatment alone). Figure 17 displays the full allocation matrix.

B.2 Aggregate Distributions

Figure 18 shows the arithmetic mean and Wasserstein barycenter side by side. Both aggregates identify medication-assisted treatment as the top priority ($\bar{P} = 0.183$, $\hat{\Psi} = 0.195$), followed by community development, medicaid expansion, and addiction treatment in prisons. The two aggregates agree on the ranking of the top interventions but differ systematically in their category-level allocation.

Table 7 reports the total budget share assigned to each category. The barycenter allocates more to treatment (0.481 vs. 0.468) and harm reduction (0.173 vs. 0.155) and less to

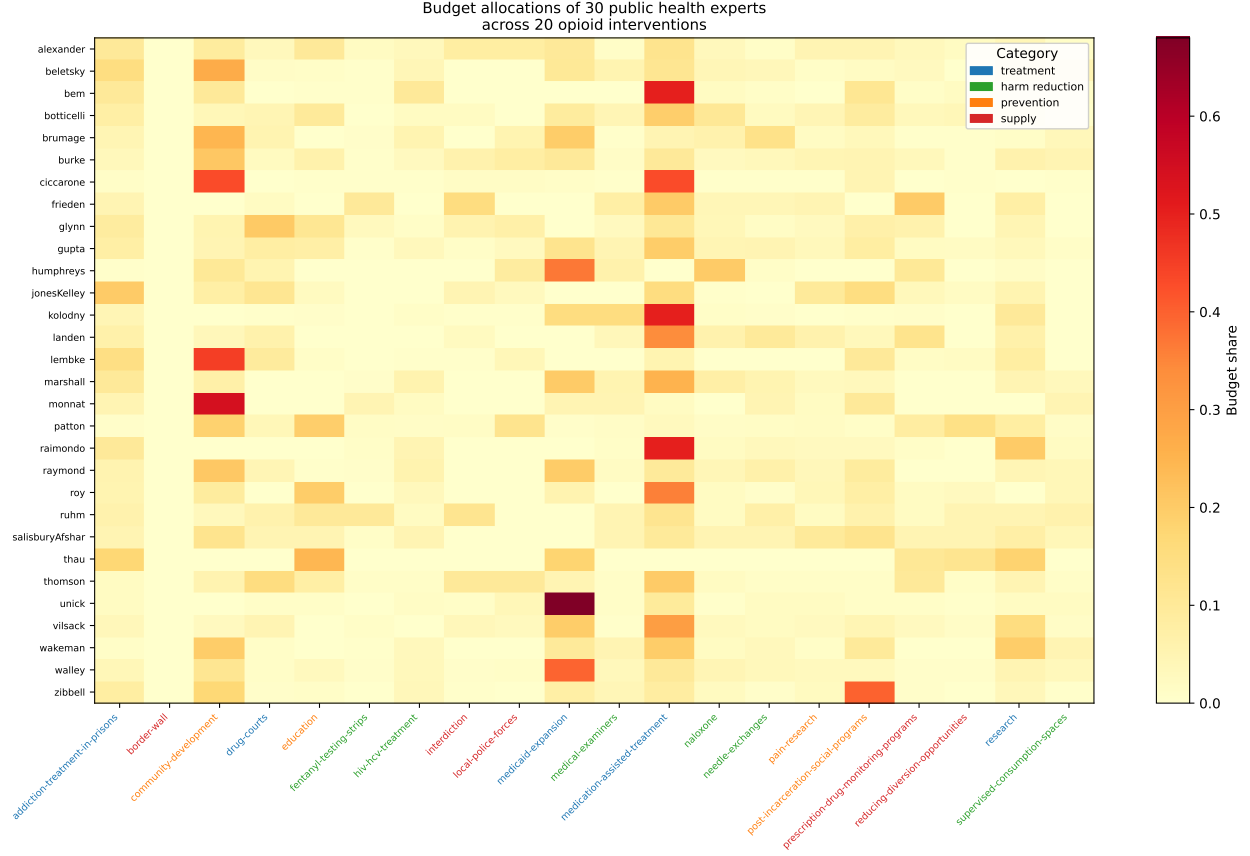


Figure 18: Budget allocations of 30 public health experts across 20 opioid interventions. Cell color encodes the budget share; intervention labels are colored by policy category (blue = treatment, green = harm reduction, orange = prevention, red = supply).

prevention (0.262 vs. 0.270) and supply (0.084 vs. 0.107) than the arithmetic mean. The largest individual-intervention differences are in community development ($\bar{P} - \hat{\Psi} = +0.013$), medical aid expansion (+0.012), and medication-assisted treatment (-0.012).

B.3 Leave-One-Out Influence

To assess the influence of individual experts, we compute the ℓ^1 shift in each aggregate when each expert is removed in turn. This is the empirical counterpart of the EIF studied in Section 3. Figure 19 displays the leave-one-out influence of each expert, ordered by their influence on the arithmetic mean.

The most influential experts on the arithmetic mean are those whose allocations deviate

Category	Arithmetic Mean	Barycenter
Treatment	0.468	0.481
Harm reduction	0.155	0.173
Prevention	0.270	0.262
Supply	0.107	0.084

Table 8: Total budget share allocated to each policy category by the arithmetic mean and Wasserstein barycenter.

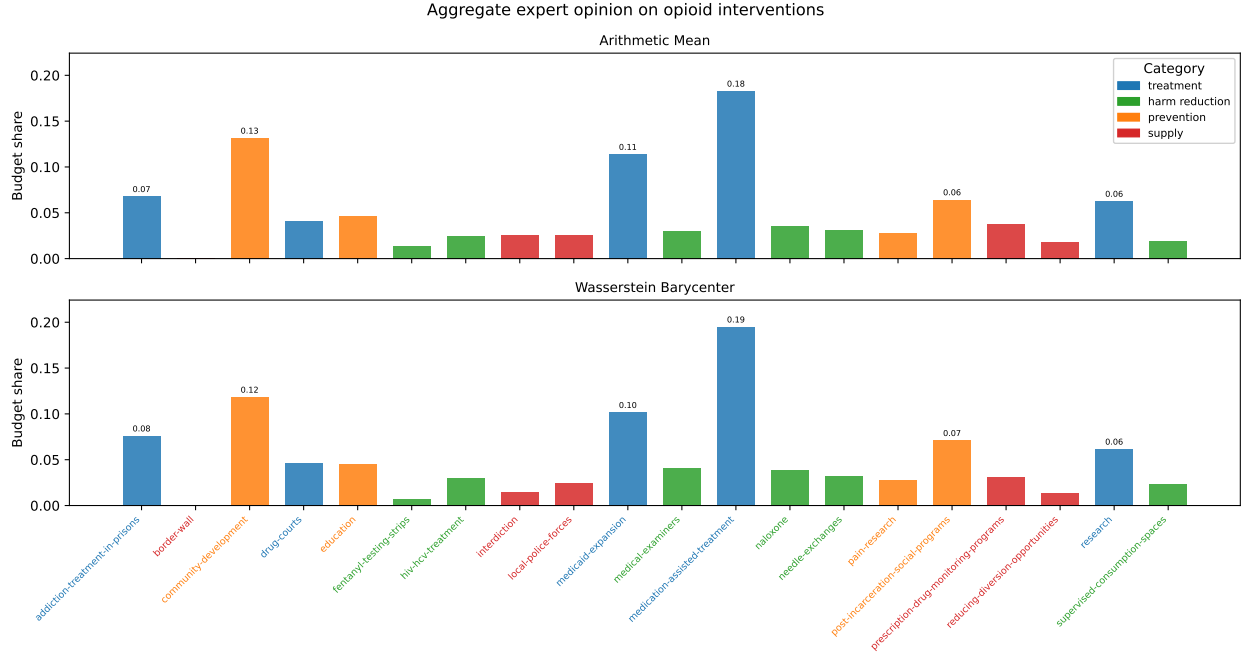


Figure 19: Arithmetic mean (top) and Wasserstein barycenter (bottom) of the 30 expert allocations. Bars are colored by policy category. Annotations show allocations exceeding 5%.

most from the consensus, concentrating nearly all of their budget on a single intervention. For this dataset, the leave-one-out shifts are comparable in magnitude for the two methods; mean shifts are 2.7 percentage points for the AM and 3.6 for the barycenter. This is consistent with the data structure: without a clearly outlying expert, both aggregators respond similarly to removals. The robustness advantage of the barycenter is most pronounced when a minority of experts holds extreme views, as in the SPF application (Section 6) and the contamination simulations (Section 4). Figure 20 further illustrates where the two methods agree and disagree at the intervention level.

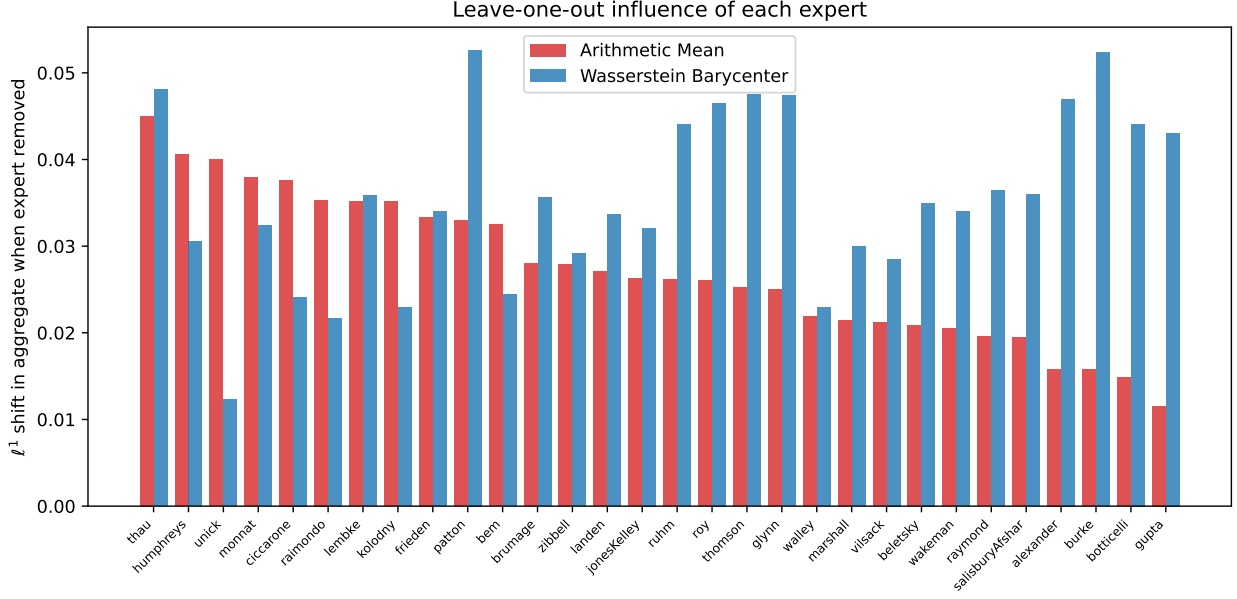


Figure 20: Leave-one-out ℓ^1 shift in the arithmetic mean (red) and Wasserstein barycenter (blue) when each expert is removed. Experts are ordered by their influence on the arithmetic mean.

C Full-Diagonal Criterion

The following proposition extends Lemma 2.1 from individual transport plans to the full barycenter solution: in any optimal solution, every transport plan simultaneously maximises its diagonal.

Proposition C.1 (Full-Diagonal Criterion). *Let $\Pi^1, \dots, \Pi^N$ be $K \times K$ matrices representing the joint distributions between a candidate barycenter Ψ and the probability mass functions $p_1, \dots, p_N$ on $\Omega = \{1, \dots, K\}$. Suppose the matrices are feasible, satisfying for all l and k*

$$\sum_j \Pi_{k,j}^l = p_l(k), \quad \sum_j \Pi_{j,k}^l = \Psi(k),$$

but for some l and i ,

$$\Pi_{i,i}^l < \min\{p_l(i), \Psi(i)\}. \quad (\text{C.1})$$

Then there exists a feasible $\tilde{\Pi}^l \neq \Pi^l$ such that $f(\tilde{\Pi}^l, \Pi^{-l}) < f(\Pi^l, \Pi^{-l})$, so $(\Pi^1, \dots, \Pi^N)$ is not optimal. Consequently, every optimal transport plan satisfies $\Pi_{kk}^l = \min\{p_l(k), \Psi(k)\}$

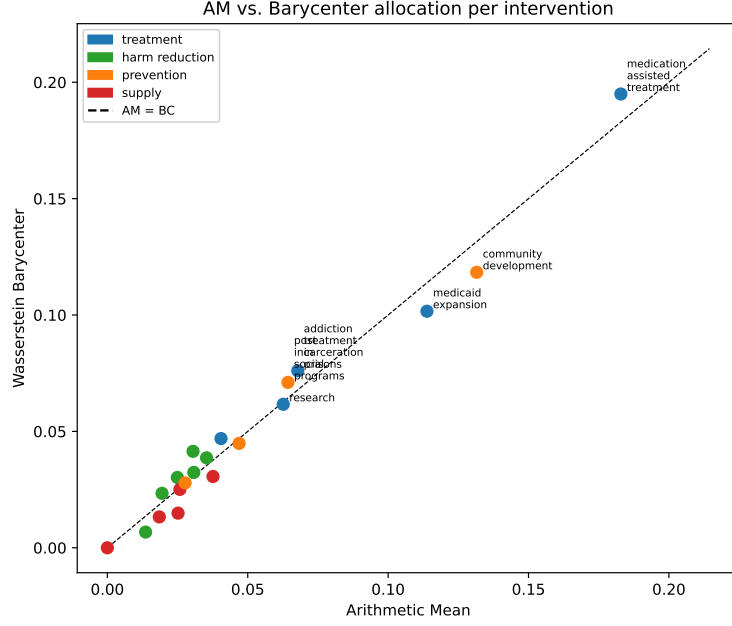


Figure 21: AM vs. barycenter allocation for each of the 20 interventions. Points above the diagonal indicate interventions where the barycenter allocates more than the AM; points below indicate the reverse. Colors denote policy category.

for all k .

Proof. The objective equals $f(\Pi) = \sum_{l=1}^N \sum_{k=1}^K (1 - \Pi_{k,k}^l)$, so increasing any diagonal entry strictly decreases the objective. Suppose (C.1) holds. Since $\Pi_{ii}^l < p_l(i)$, there exists $k_1 \neq i$ with $\Pi_{i,k_1}^l > 0$. Setting $\tilde{\Pi}_{i,i}^l = \Pi_{i,i}^l + \Pi_{i,k_1}^l$ and $\tilde{\Pi}_{i,k_1}^l = 0$ preserves row i 's sum. Since $\Pi_{ii}^l < \Psi(i)$, the column- i sum $\Psi(i)$ exceeds the diagonal; there exists a set of indices $\{k_2^i\}$ and weights $\{\epsilon_i\}$ summing to Π_{i,k_1}^l such that transferring mass $\epsilon_i \Pi_{k_2^i,i}^l$ from column i to column k_1 restores all column sums. The resulting $\tilde{\Pi}^l$ is feasible and the objective decreases by $\Pi_{i,k_1}^l > 0$. $\square$

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